

EN3150 Pattern Recognition
Linear Models for Regression

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Semester 5 – Batch 21.

Notation

- The vectors are denoted by boldface lower-case letter. e.g., $\boldsymbol{a}$
- Vectors written with commas represent column vectors

$$\boldsymbol{v} = [a,\ b,\ c],\ \boldsymbol{v} = [a\ b\ c]^T,\ \text{or}\ \boldsymbol{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$[a,\ b,\ c]^T \equiv [a\ b\ c]^T$$

- The matrices are denoted by boldface upper-case letter, e.g., $\boldsymbol{A}$

$$\boldsymbol{a} = [a_1\ a_2\ \dots a_i \dots a_n\ ]^{\boldsymbol{T}} \qquad \boldsymbol{b} = [b_1\ b_2\ \dots b_i \dots b_n\ ]^{\boldsymbol{T}}$$

$$\boldsymbol{a^T\ b} = \underset{1 \text{ x } n}{[a_1\ a_2\ \dots a_i \dots a_n\ ]} \underset{n \text{ x } 1}{\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \boldsymbol{b^T\ a}$$

$\boldsymbol{ab}$?

n x 1 n x 1

$\boldsymbol{ab^T}$?

n x 1 1 x n

$$\begin{bmatrix} a_1 b_1 & \cdots & a_1 b_n \\ \vdots & \ddots & \vdots \\ a_n b_1 & \cdots & a_n b_n \end{bmatrix}$$

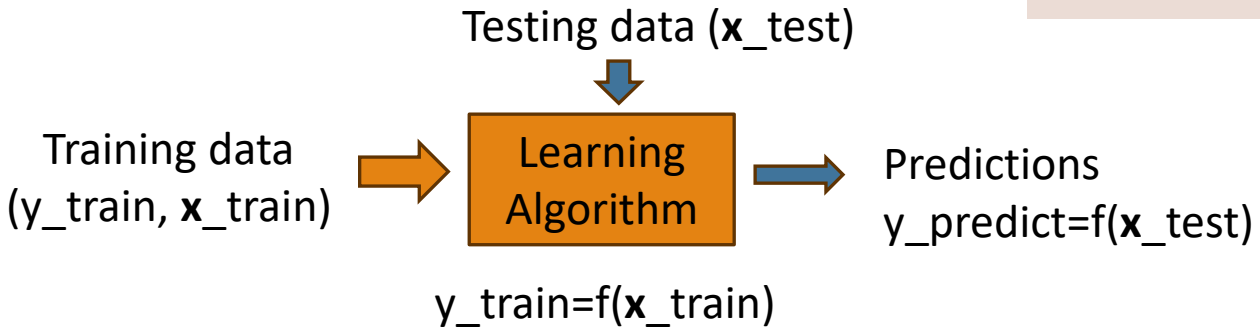
n x n Matrix

What is regression ?

Predict the value of **dependent variable** based on the values of one or more **independent variables**

Dependent variable is often a quantitative or continuous variable

- Regression is a supervised learning technique



- Applications
  - Trend forecasting (e.g., forecast future sales)
  - Forecasting an effect (e.g., predicting the effect of advertising spending on sales)
  - To determine the strength of predictors (regression coefficients can indicate the magnitude and direction of influence each predictor)

linear

Key questions

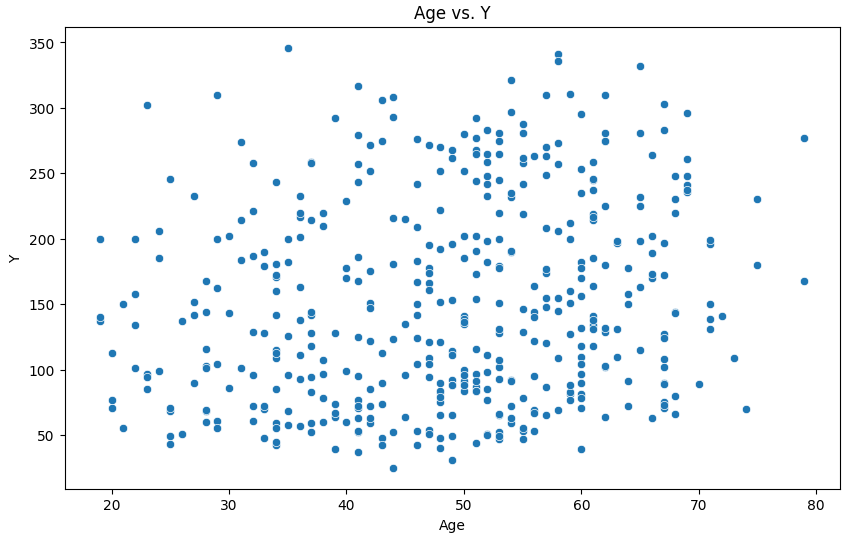
- Is there a relationship between dependent variable and independent variable(s)?
- How strong is the relationship ? and is it positive or negative relationship ? Linear or non-linear?
- How accurate our predictions?
- Is there synergy among features/ independent variables

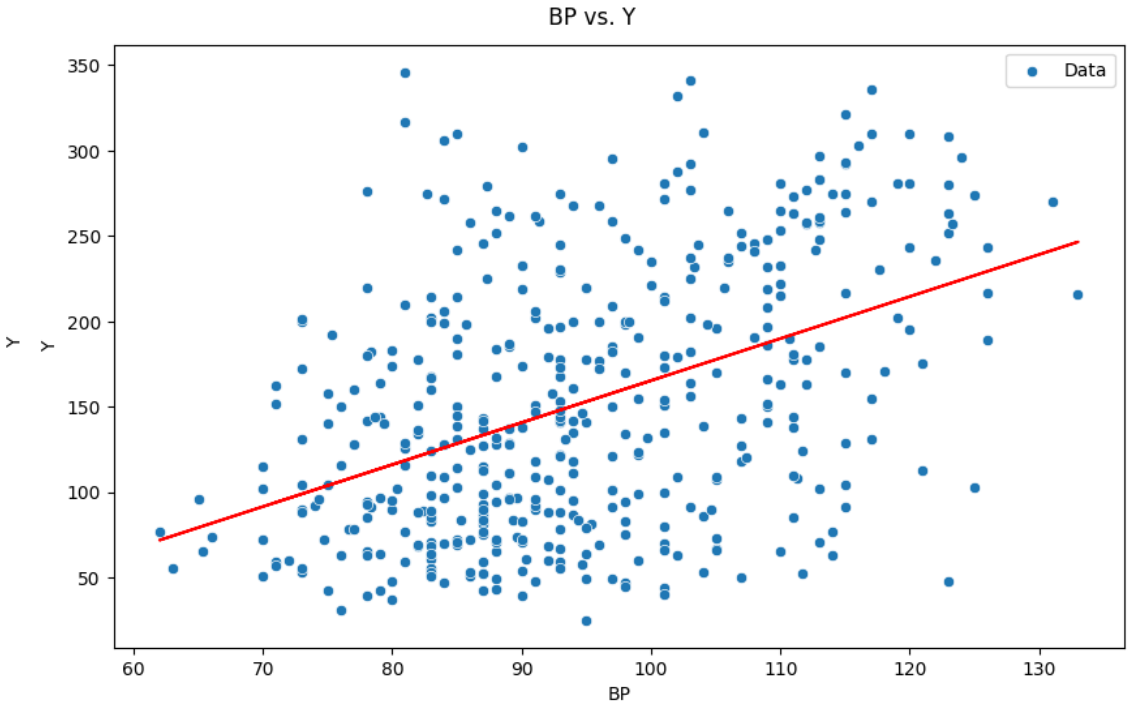
Diabetes dataset: independent variable(s) Diabetes Data (ncsu.edu)	
age	tc total serum cholesterol
sex	ldl, low-density lipoproteins
body mass index	hdl, high-density lipoproteins
average blood pressure	ltg, possibly log of serum triglycerides level
tch, total cholesterol / HDL	glu, blood sugar level

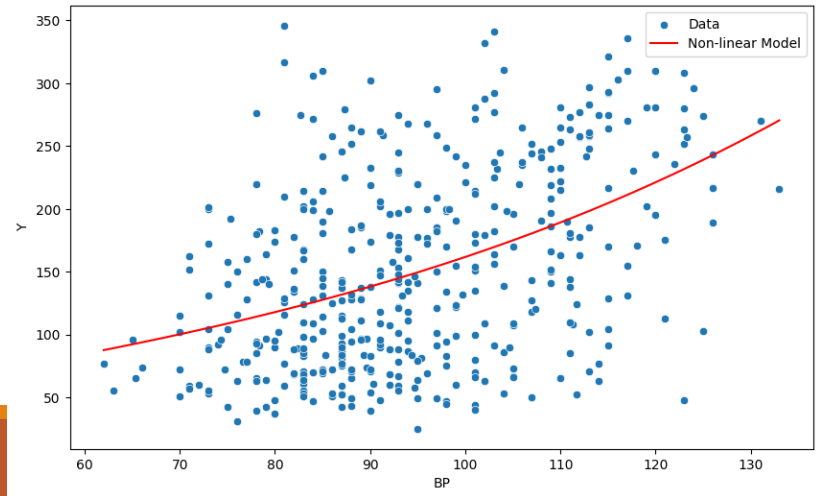
Dependent variable: quantitative measure of disease progression one year after baseline

Example: **Diabetes dataset**

	AGE	SEX	BMI	BP	S1	S2	S3	S4	S5	S6	Y
0	59	2	32.1	101.0	157	93.2	38.0	4.0	4.8598	87	151
1	48	1	21.6	87.0	183	103.2	70.0	3.0	3.8918	69	75
2	72	2	30.5	93.0	156	93.6	41.0	4.0	4.6728	85	141
3	24	1	25.3	84.0	198	131.4	40.0	5.0	4.8903	89	206
4	50	1	23.0	101.0	192	125.4	52.0	4.0	4.2905	80	135

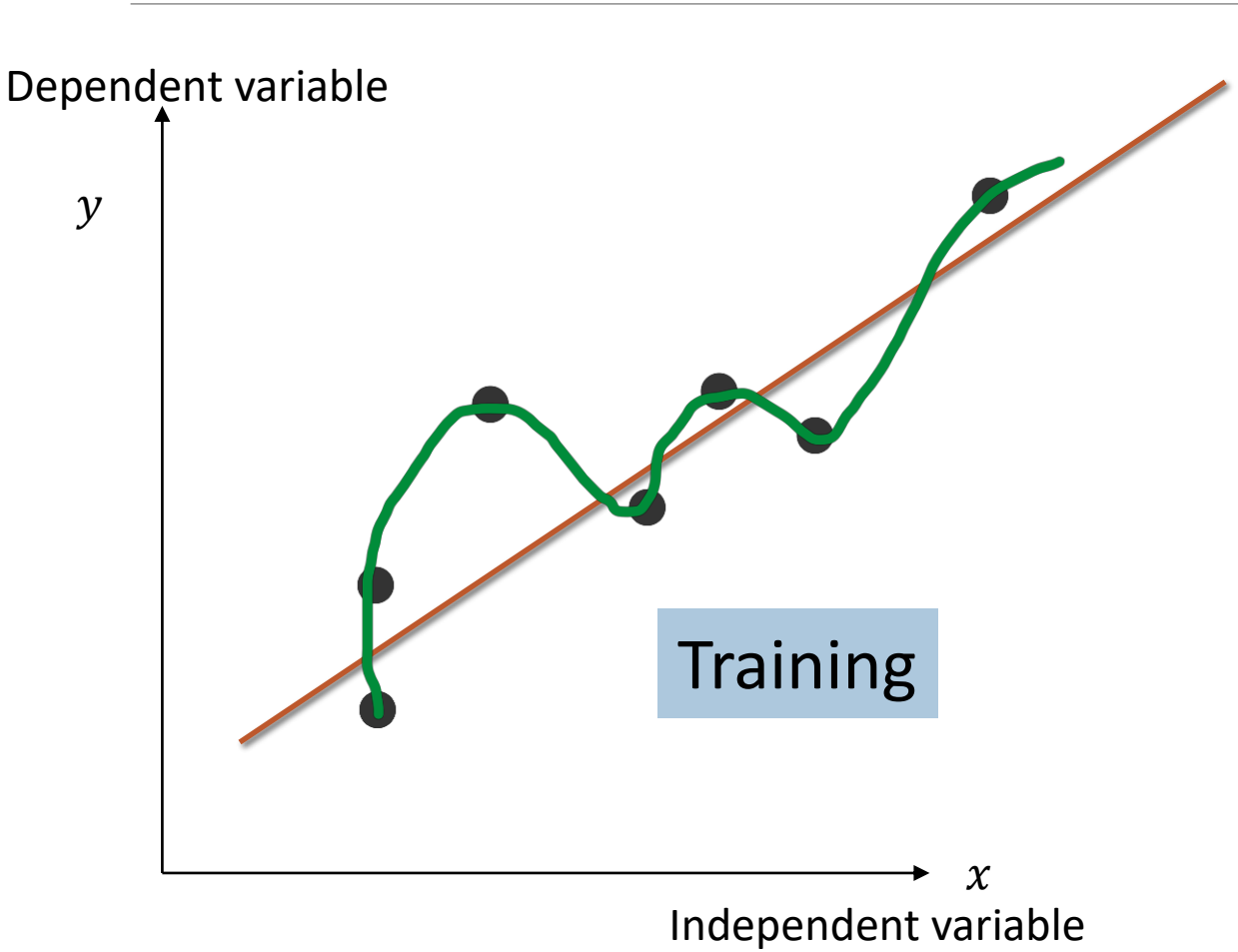


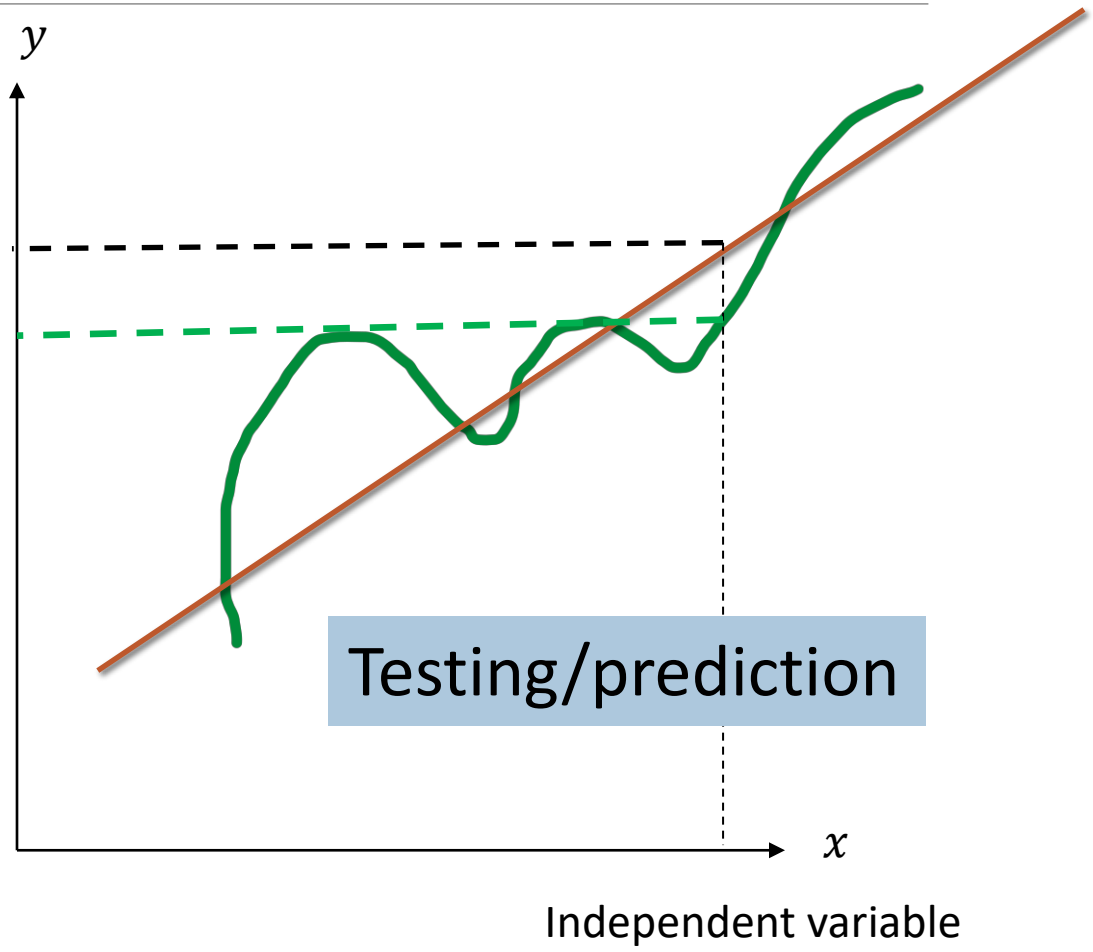




[Diabetes Data (ncsu.edu)]

What is regression ?





Linear Regression

- Most basic form of regression
- Relationship between the dependent and independent variables is assumed to be linear

assumed to be linear

$$y(\mathbf{x}) \approx w_0 + w_1 x_1 + \cdots, + w_D x_D$$
$$y(\mathbf{x}) = w_0 + w_1 x_1 + \cdots, + w_D x_D + \epsilon = \mathbf{w}^T \mathbf{x} + \epsilon$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ . \\ . \\ . \\ x_D \end{bmatrix} \qquad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ . \\ . \\ . \\ w_D \end{bmatrix}$$

- w_0 - Intercept or bias term
- What bias represents? ⟶ what is the output when all features are zero
- ϵ – Error term, Why this term? ⟶ measurement, internal errors.
- $\mathbf{w}_{(1:D)}$ – Weights or regression coefficients **(parameters)** of independent variables

Linear Regression

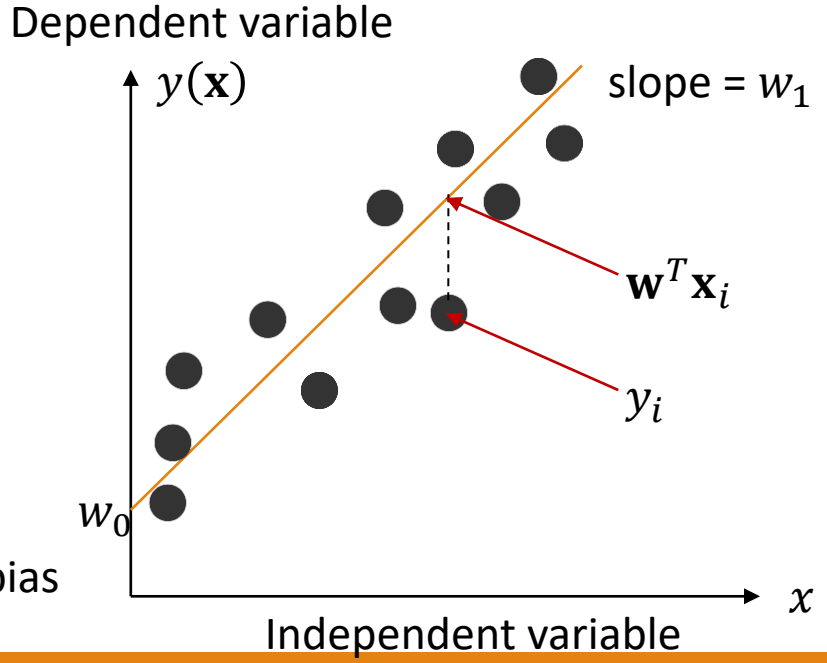
$$y(\mathbf{x}) = w_0 + w_1 x_1 + \cdots, + w_D x_D + \epsilon = \mathbf{w}^T\mathbf{x} + \epsilon$$

$$\mathbf{y}=\begin{bmatrix} y_1 \\ . \\ y_i \\ . \\ y_N \end{bmatrix} \quad \mathbf{w}=\begin{bmatrix} w_0 \\ w_1 \\ . \\ . \\ w_D \end{bmatrix} \quad \mathbf{x}_i=\begin{bmatrix} 1 \\ x_{1,i} \\ . \\ . \\ x_{D,i} \end{bmatrix}$$

- $\mathbf{w}^T\mathbf{x}$ = inner or scalar product between the input vector $\mathbf{x}$ and the weight vector $\mathbf{w}$
- $y(\mathbf{x})$ is dependent variable (continuous)
- $\mathbf{w}$ To be learned from data. How?
- Cost function

$$\text{J}(\mathbf{w}) = \sum_{i=1}^{N} (y_i - \mathbf{w}^T\mathbf{x}_i)^2$$

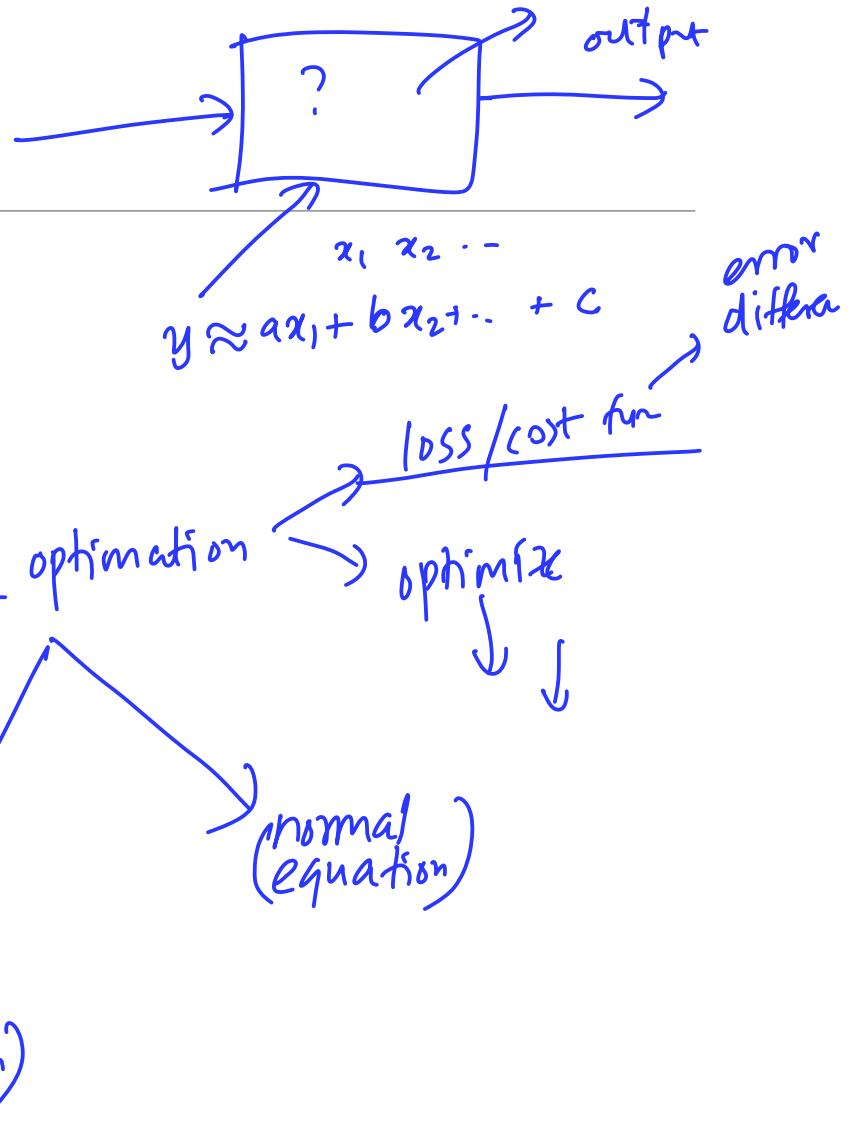
- Objective: Find $\mathbf{w}$ which has the minimum J($\mathbf{w}$) for our training samples (N number of samples)
- How?



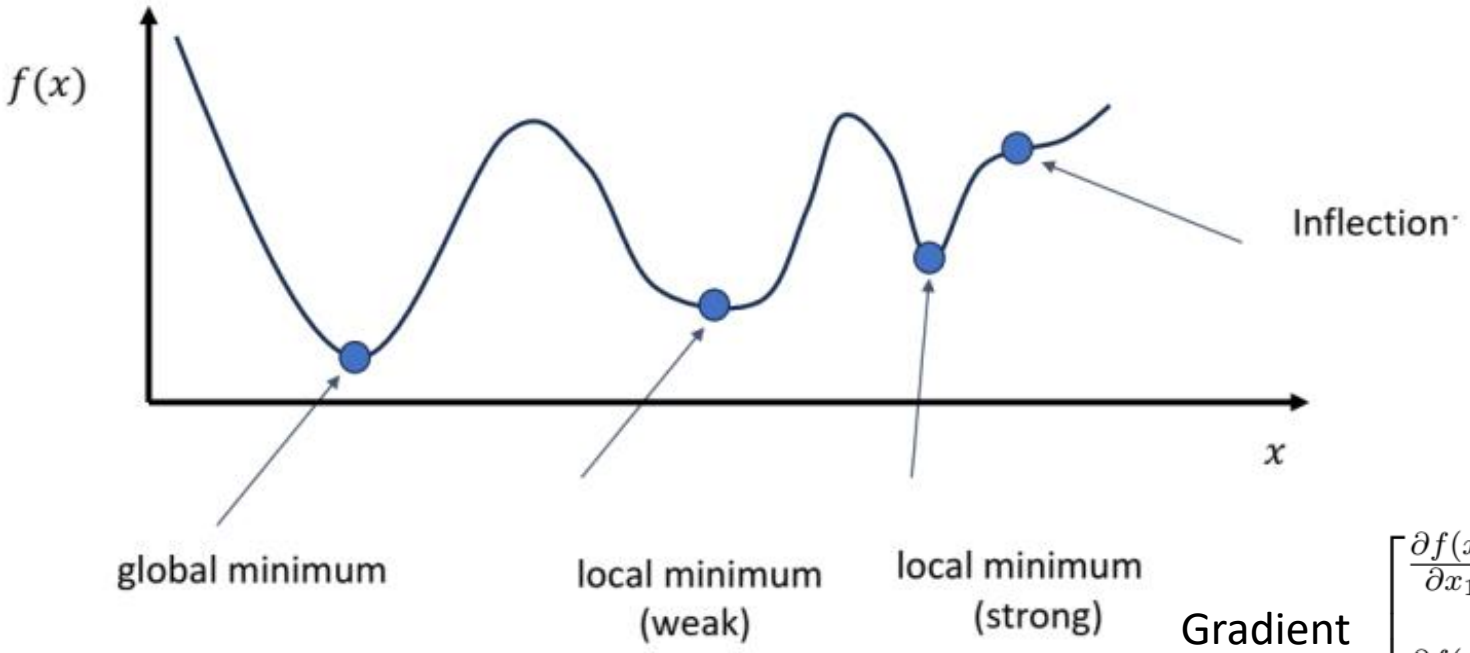
Linear Regression

- $y(\mathbf{x}) = w_0 + w_1 x_1 + \cdots, + w_D x_D + \epsilon = \mathbf{w}^T \mathbf{x} + \epsilon$
- Hypothesis: $y(\mathbf{x}) = w_0 + w_1 x_1 + \cdots, + w_D x_D$
- Parameters : $w_0, w_1, \ldots, w_D$
- Cost function : $J(\mathbf{w}) = \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)^2$
- Objective: Find $\mathbf{w}$ which has the minimum $J(\mathbf{w})$

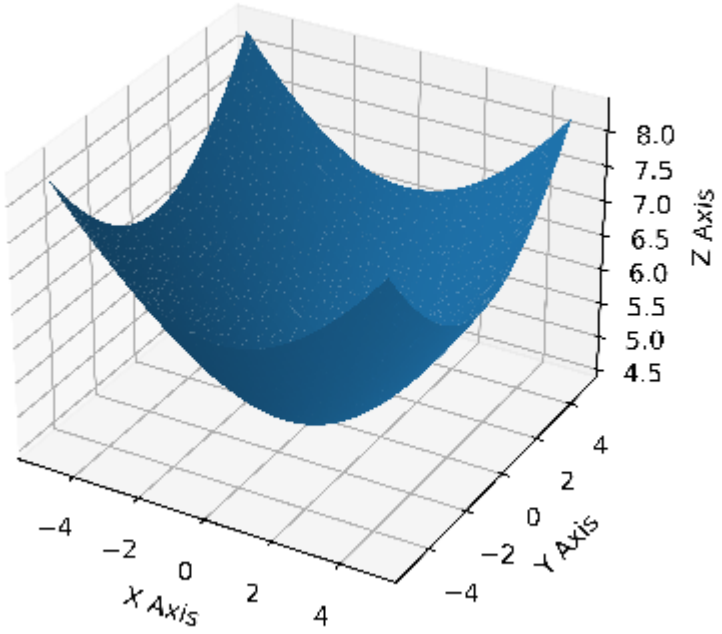
$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} J(\mathbf{w})$$



Linear Regression

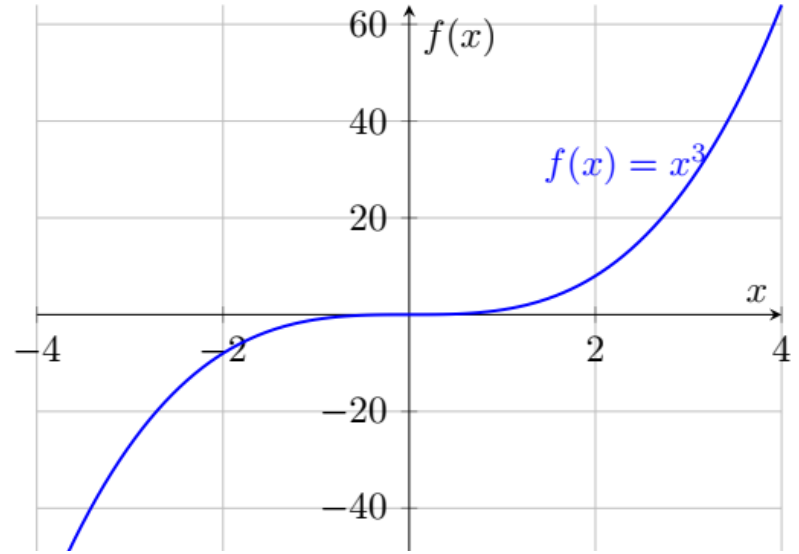


$\sqrt{x^2 + y^2 + 20}$



Gradient

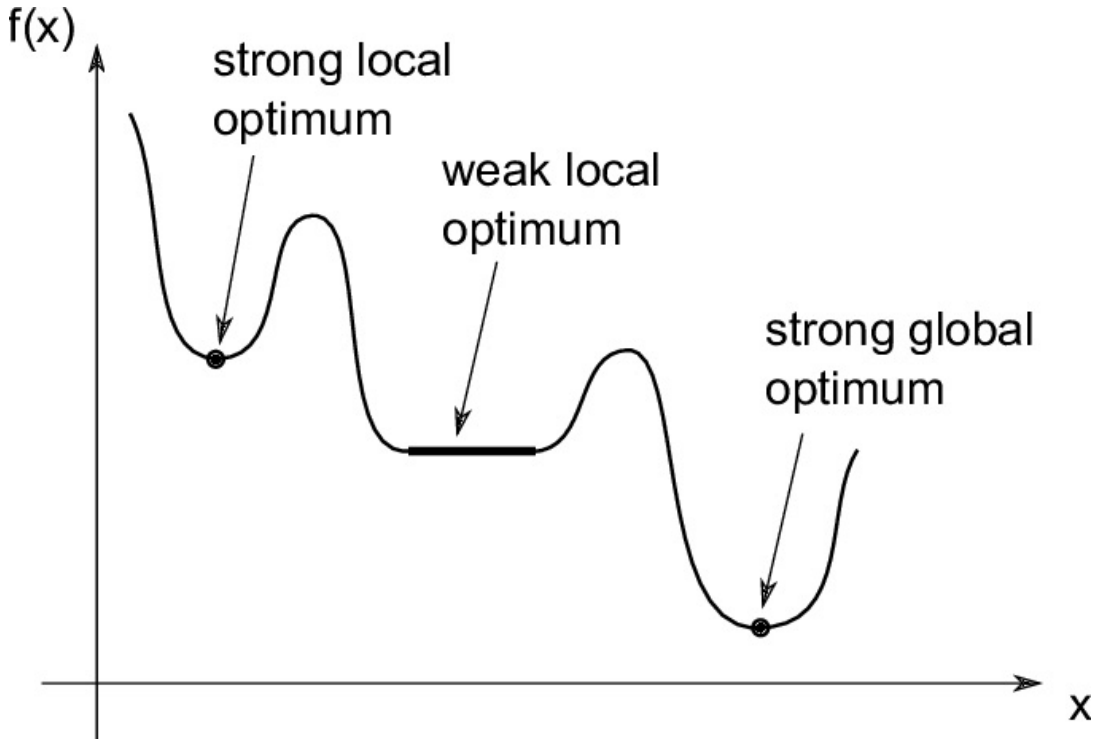
$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$



First-order necessary condition (FONC) $\nabla f(\boldsymbol{x}) = 0$

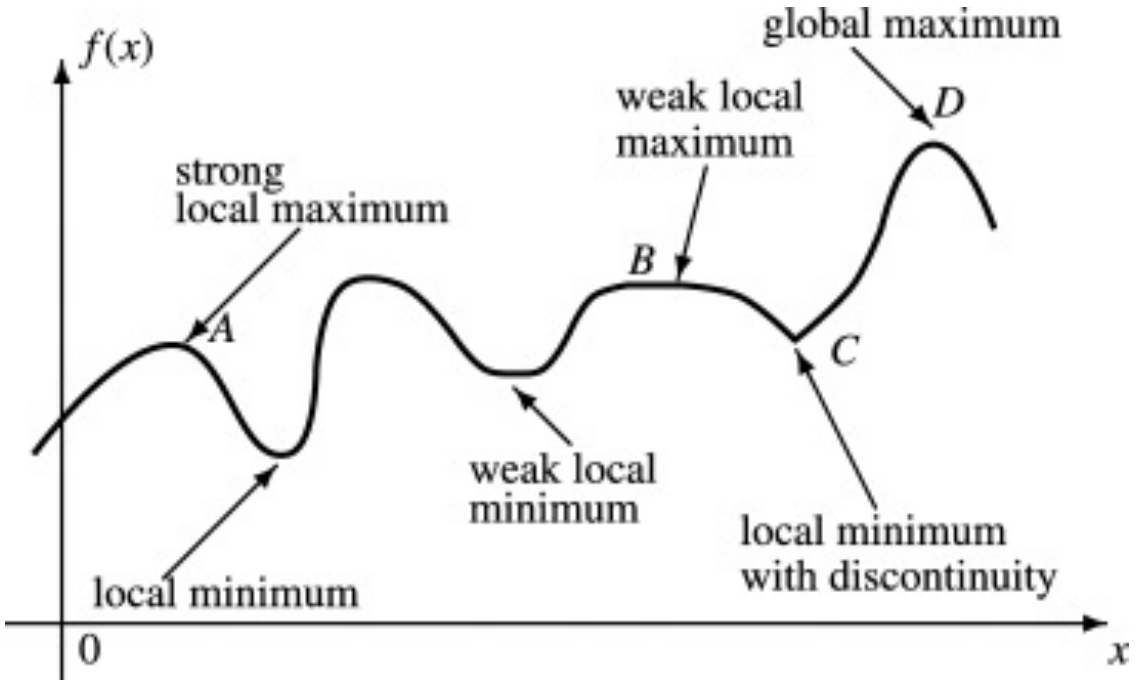
used to identify possible candidates, but is not sufficient

$f'(x) = 0$, but $x = 0$ is not a minimum or maximum

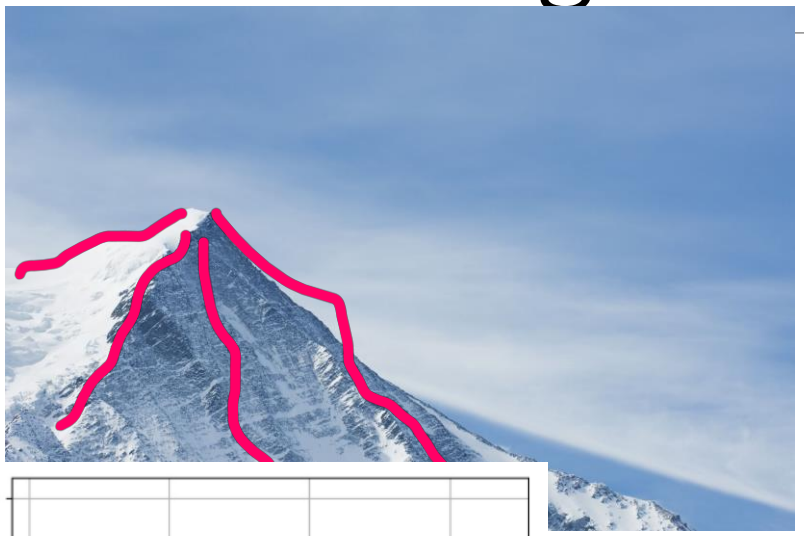


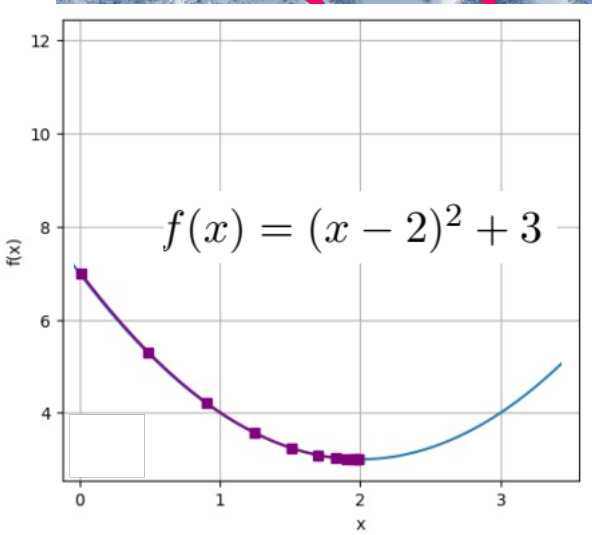
* Strong optima converge faster
* less likely suboptimal
* more stable

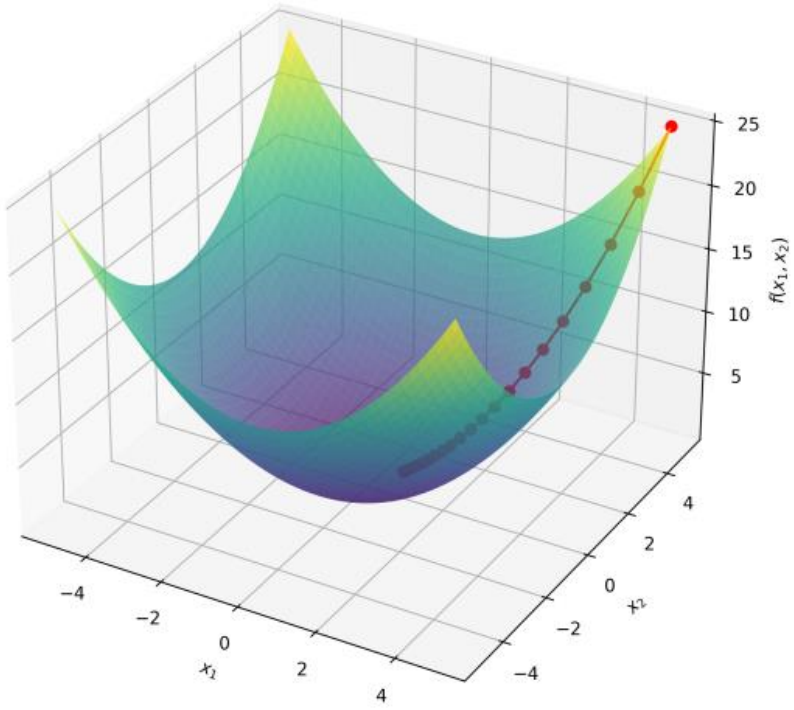
* weak local optima are slower to converge
* plateaus with zero gradients cause optimizing problem to slow down and stall



Linear Regression





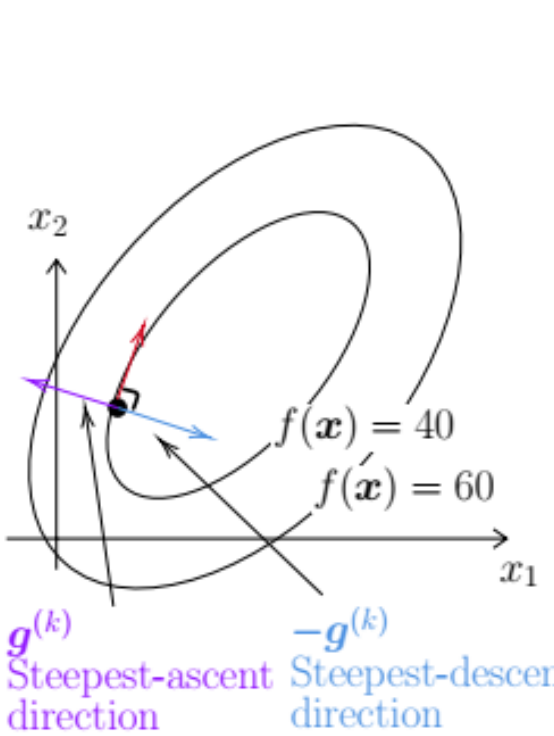


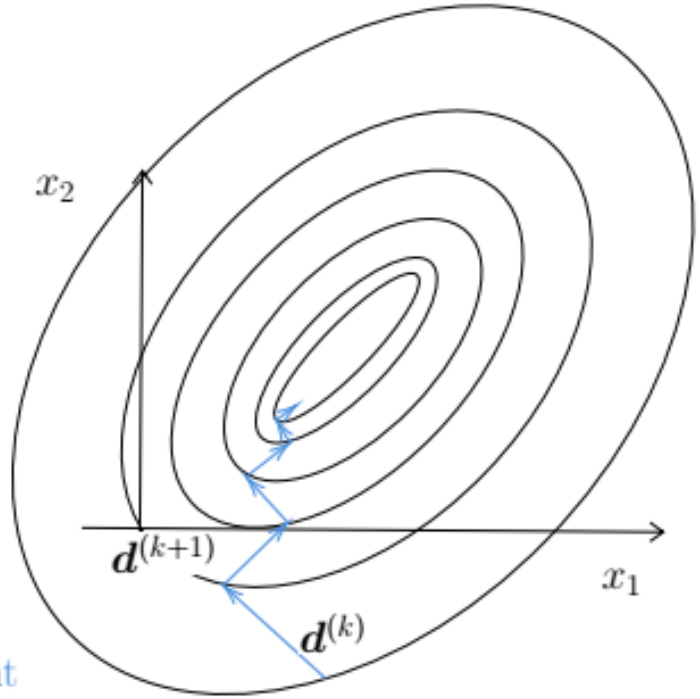
$$f(\boldsymbol{x}) = \frac{1}{2}(x_1^2 + bx_2^2) \text{ with } b = 1,\ \boldsymbol{x}^{(0)} = [5, 5]$$

What is Gradient Descent? | IBM

Linear Regression







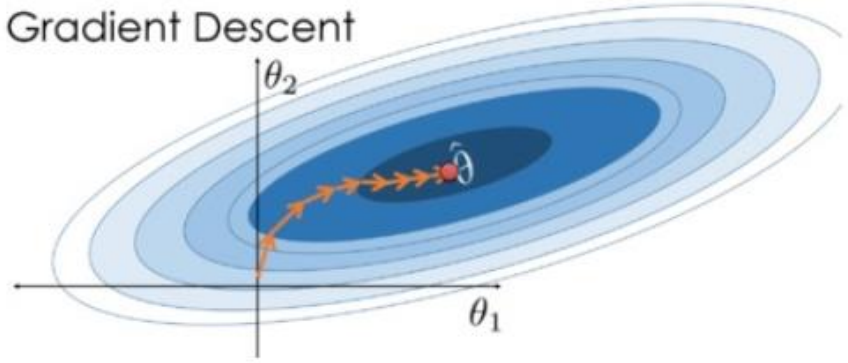


Image is adapted from Lau, S., Gonzalez, J., Nolan, D. (2020) https://www.textbook.ds100.org

$w^T = [w_0\ w_1 \cdots w_D]$ $x_i = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}$ y_i is a scalar

Linear Regression



- Cost function → $J(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2$

this is a scalar

- Objective: Find which has the minimum J($\mathbf{w}$) for our training samples (N number of samples)
- $\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} J(\mathbf{w})$

Gradient descent

learning rate

Update for w_j $$w_j \leftarrow w_j - \alpha \frac{\partial\ J(\mathbf{w})}{\partial w_j}$$

$$y_i = w_0 + w_1\, x_{1,i} + \cdots, + w_D\, x_{D,i}$$

For one training sample

$$\frac{\partial\ J(\mathbf{w})}{\partial w_j} = \frac{1}{2}\frac{\partial(y_i - \mathbf{w}^T\mathbf{x}_i)^2}{\partial w_j} = \frac{2(y_i - \mathbf{w}^T\mathbf{x}_i)}{2}\frac{\partial(y_i - \mathbf{w}^T\mathbf{x}_i)}{\partial w_j} = (y_i - \mathbf{w}^T\mathbf{x}_i)(-x_{j,i})$$

If model output and true value are close, we need little update on parameter

Update rule

$$w_j \leftarrow w_j + \alpha \cdot \sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)(x_{j,i})$$

Batch gradient descent

What is Gradient Descent? | IBM

Gradient Descent Explained (ibm.com)

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \cdot \\ y_i \\ \cdot \\ y_N \end{bmatrix} \qquad \boldsymbol{X} = \begin{bmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & x_{1,N} & x_{D,N} \end{bmatrix} \qquad w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_D \end{bmatrix}$$

$x_{j,i}$: feature number j, example number i

$$Xw = \begin{pmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ & & \\ 1 & x_{1,N} & x_{D,N} \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_D \end{pmatrix} = \begin{pmatrix} w_0 + w_1 x_{1,1} + \cdots + w_D x_{D,1} \\ w_0 + w_1 x_{1,2} + \cdots + w_D x_{D,2} \\ \\ w_0 + w_1 x_{1,D} + \cdots + w_D x_{D,D} \end{pmatrix} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_N \end{pmatrix}$$

1st example; 1st feature; D^{th} feature; N^{th} example

$$y - Xw = \begin{pmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \\ y_N - \hat{y}_N \end{pmatrix}$$

$$\| y - Xw \|^2 = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \cdots + (y_N - \hat{y}_N)^2$$

$$\| y - Xw \|^2 = (y - Xw)^T (y - Xw)$$

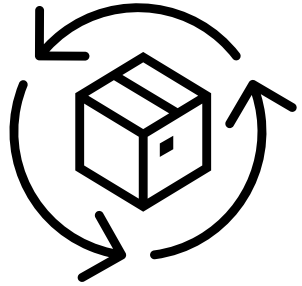
Update rule

$$w := w + \alpha X^T (y - Xw)$$

w: $D \times 1$; X^T: $D \times N$; $(y - Xw)$: $N \times 1$

* vectorized implementation is more computationally efficient

Linear Regression



Initialize $\mathbf{w}$

Repeat until convergence {

$$w_j^{(k+1)} \leftarrow w_j^{(k)} + \alpha \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)(x_{j,i}) \qquad \text{for every j}$$

}

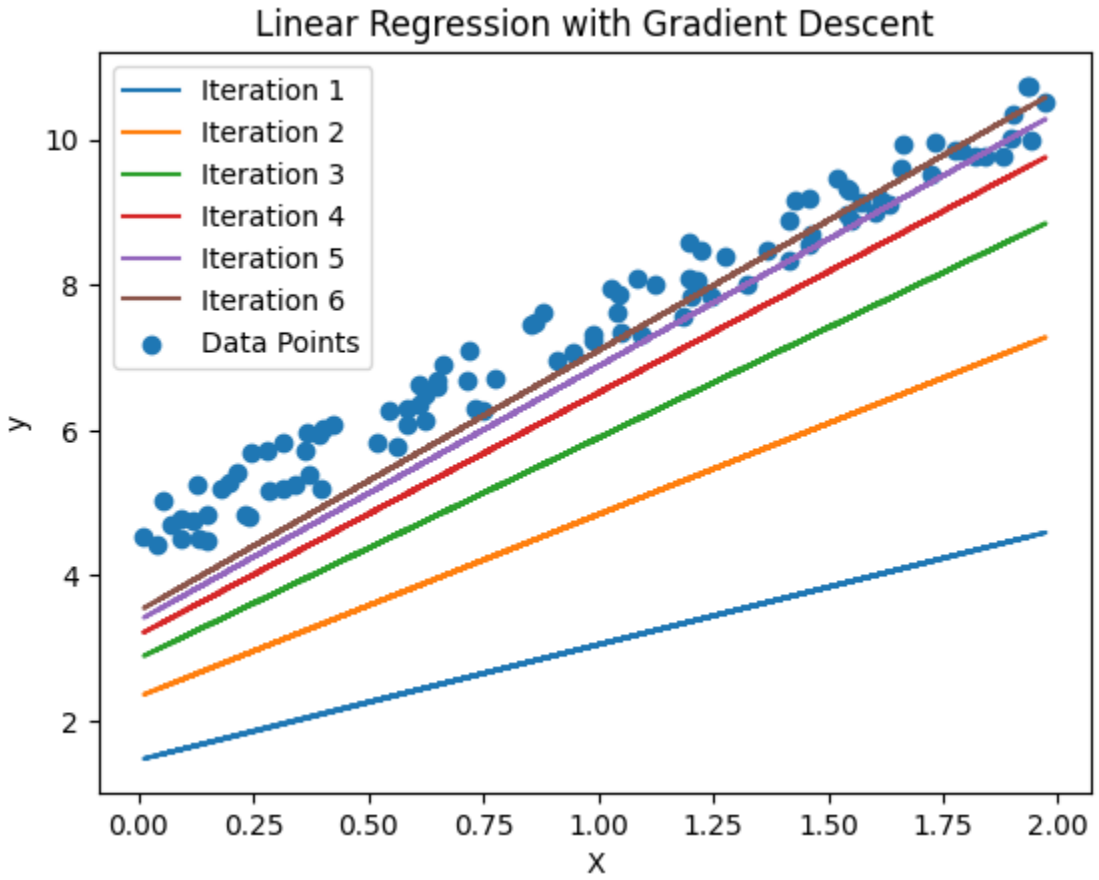
iterative algorithm

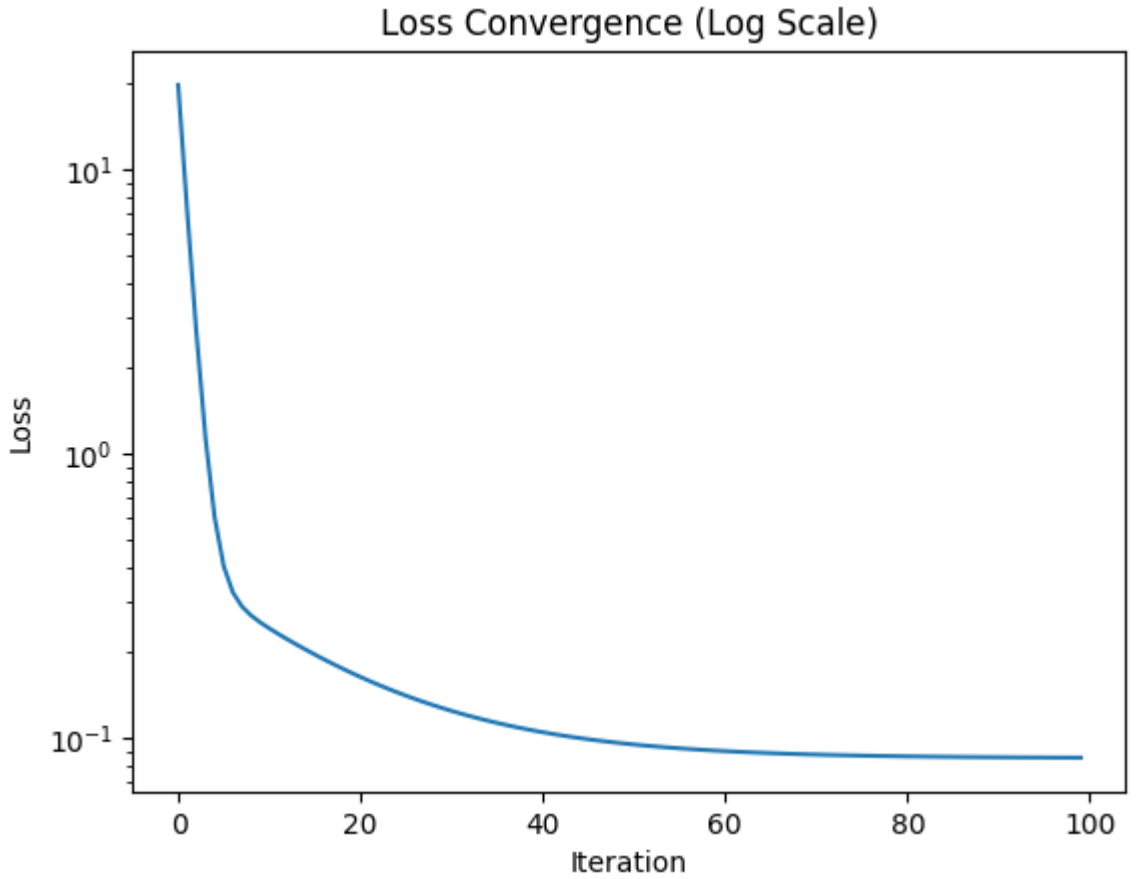
Weight update for the (k+1) th iteration

$y = f(x) + \varepsilon$

Linear Regression

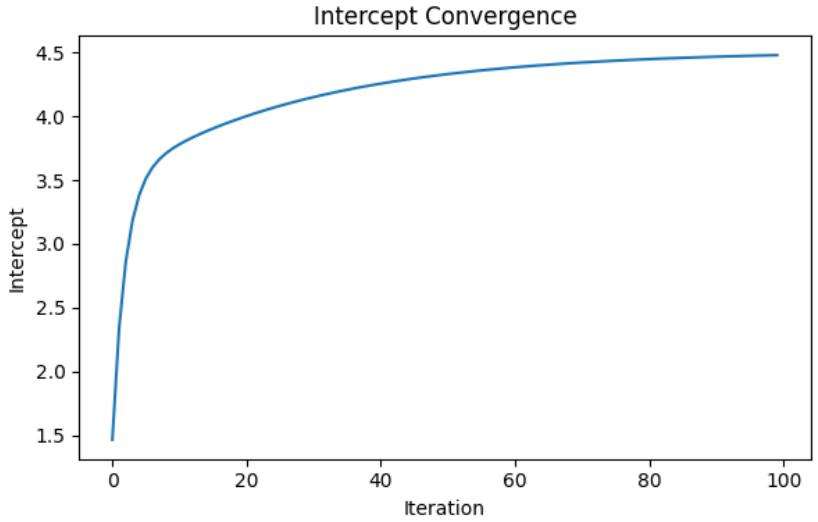
y= 4 + 3x, Initialization w_0, w_1=[0,0], learning rate =0.001

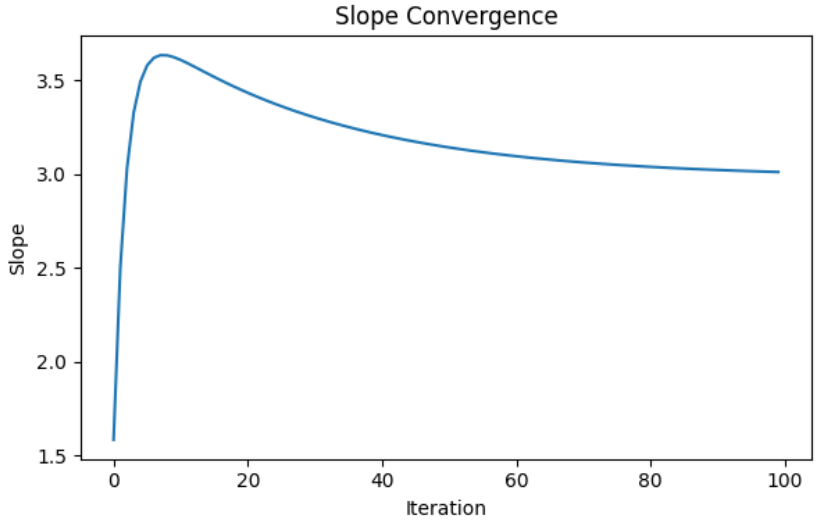




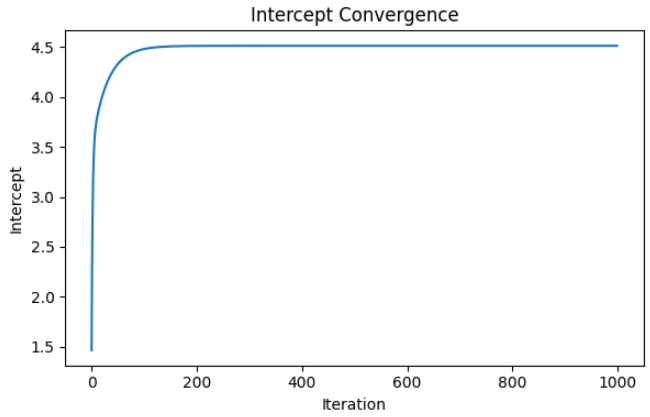
Linear Regression

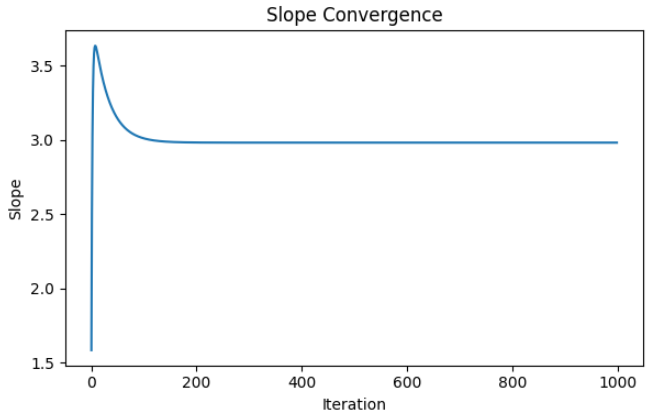
y= 4 + 3x, Initialization $w_{0,}w_1$=[0,0], learning rate =0.001





After 100 iterations
Intercept: 4.48
Slope: 3.01

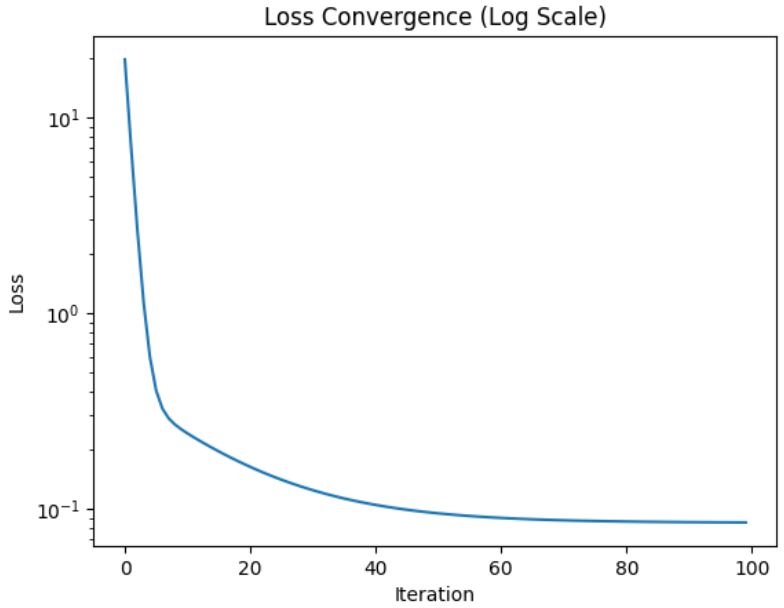




Linear Regression

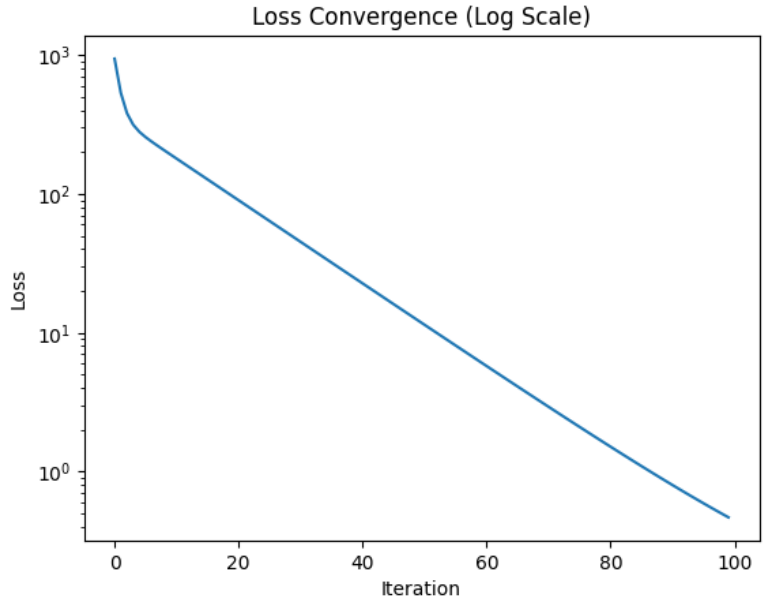
y= 4 + 3x, learning rate =0.001

Initialization w_0, w_1=[0,0],



After 100 iterations
Intercept: 4.48
Slope: 3.01

Initialization w_0, w_1=[-50,13],

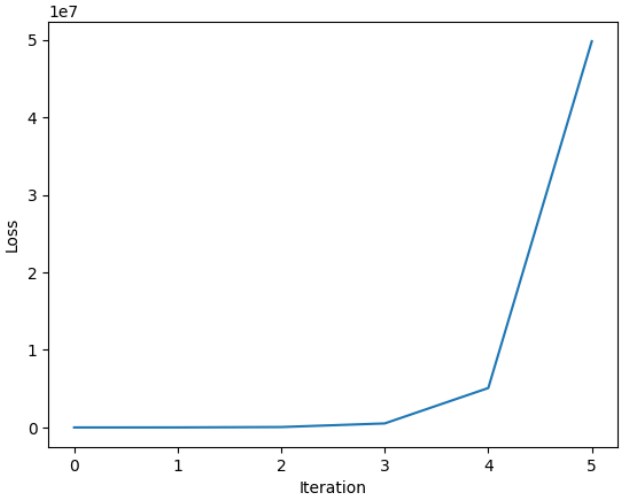


After 100 iterations
Intercept: 3.38
Slope: 3.97

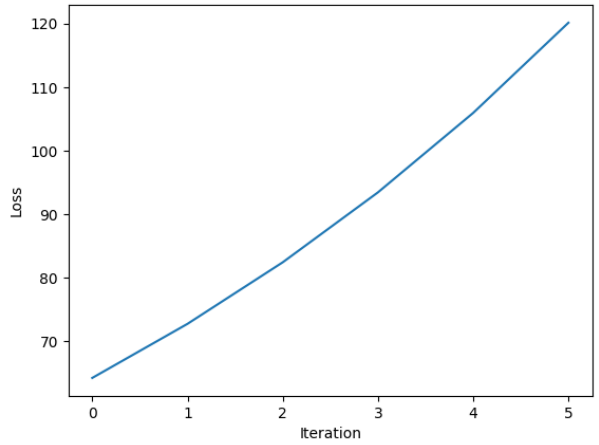
Linear Regression

Y= 4 + 3x, Initialization $w_{0,}w_1$=[0,0],

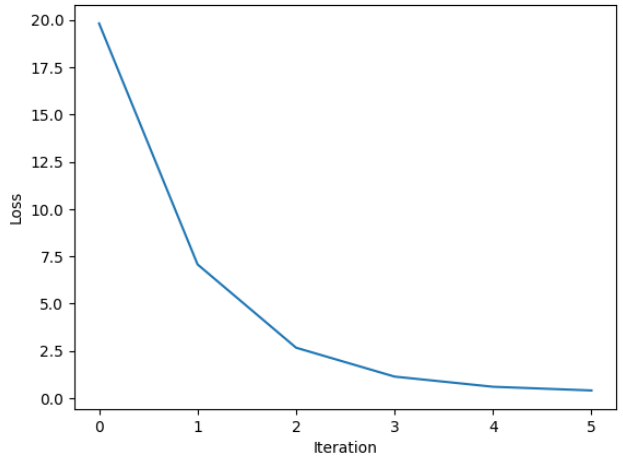
learning rate =0.01



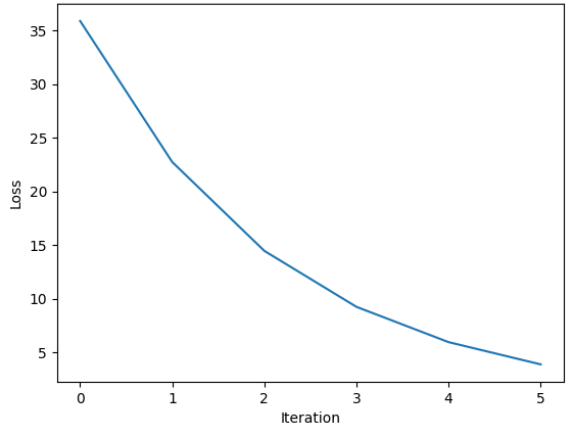
learning rate =0.005



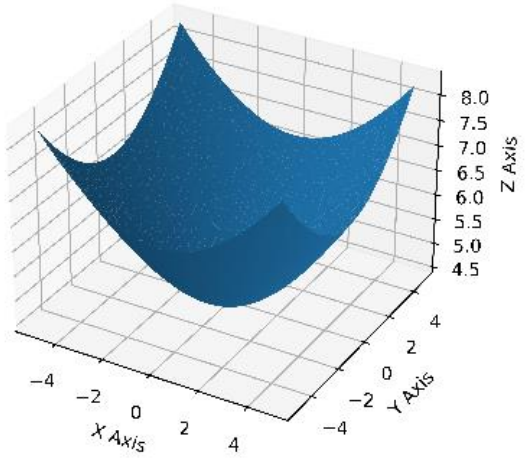
learning rate =0.001



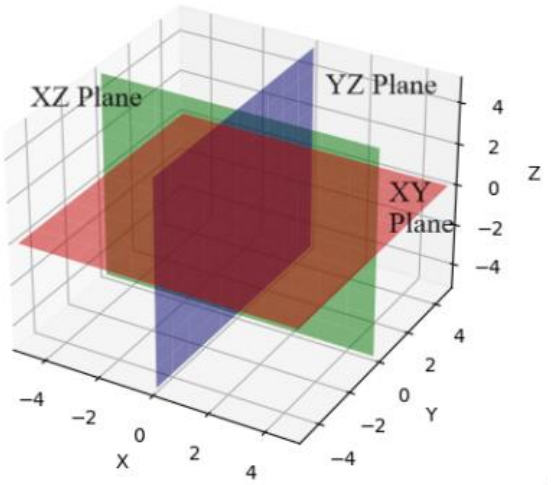
learning rate =0.0005



Linear Regression



(a) $\sqrt{x^2 + y^2 + 20}$



(b) Coordinate planes

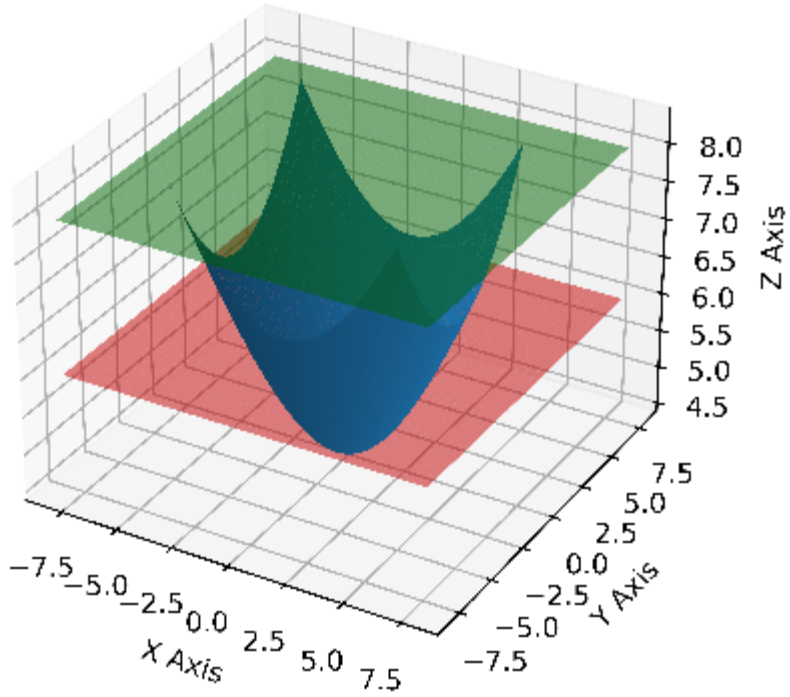


Figure: $\sqrt{x^2 + y^2 + 20}$

Suppose you are looking at XY plane

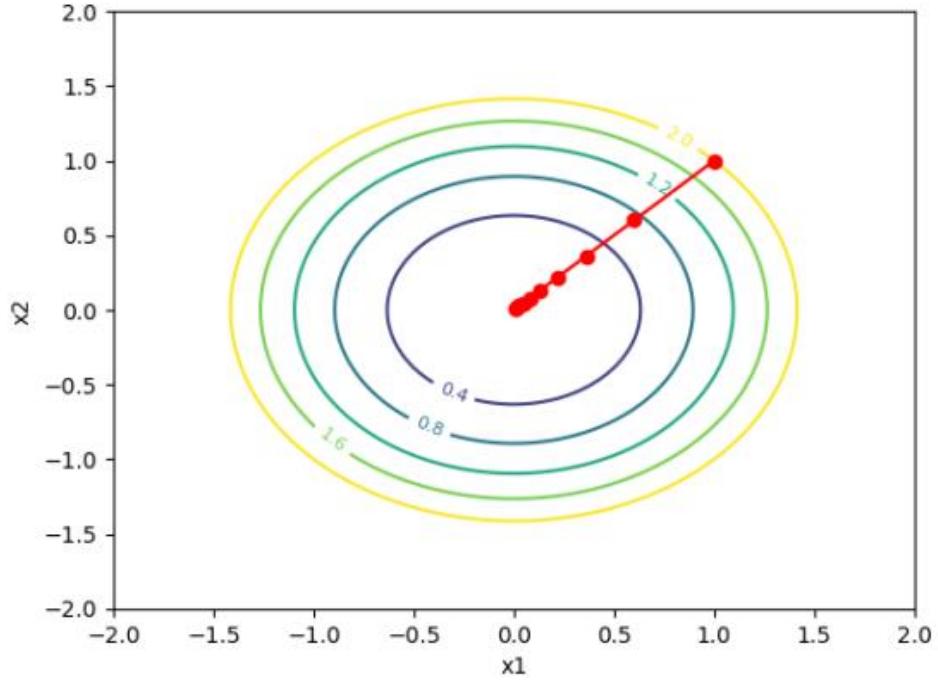
What will you see?

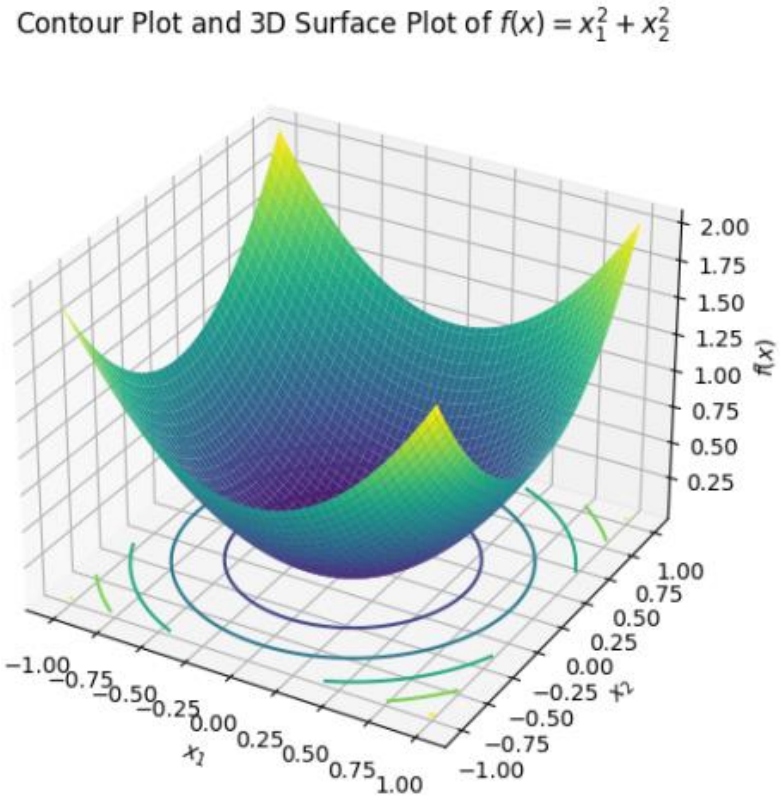
Why is it important?

A contour plot is a graphical representation used to visualize a function F(x,y) of two variables. In this type of plot, distinct colors are assigned to different values of F. You'll observe a set of curves on the graph. These curves are called contours and are drawn in such a way that they follow paths along which the values of F(x,y) remain constant. Each contour line corresponds to a specific value of F.

Linear Regression

$$f(\boldsymbol{x}) = x_1^2 + x_2^2$$





Gradient descent variants

Batch Gradient Descent

- Calculate the gradient (derivative) of the objective function with respect to all the training data points
- Slow but more accurate

Stochastic Gradient Descent (SGD)

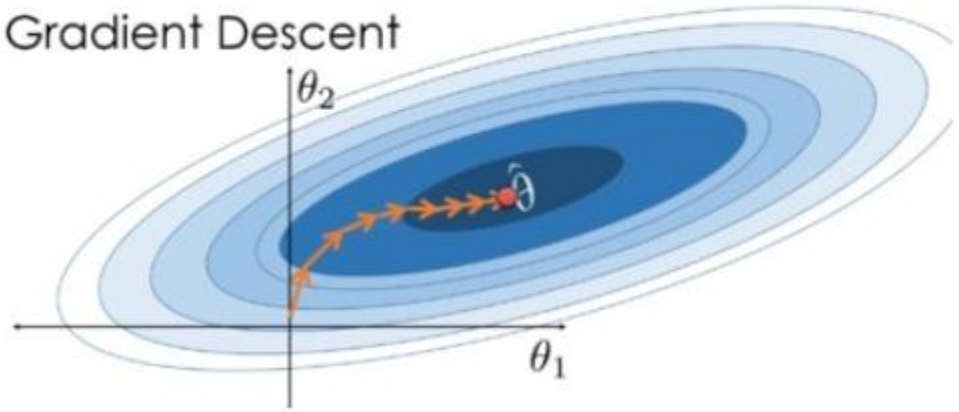
- SGD uses only one randomly selected data point at a time
- Faster convergence
- Can be noisy and might converge to a suboptimal solution

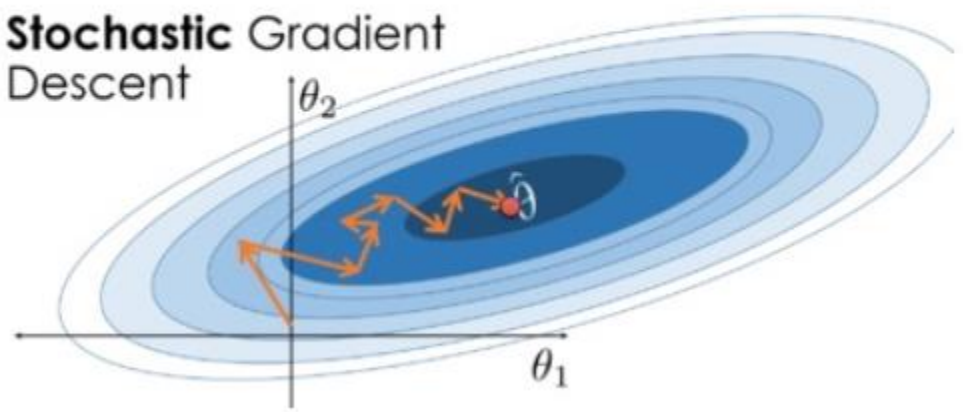
Mini-batch Gradient Descent:

- Computes the gradient using a small randomly selected subset (mini-batch) of the dataset

Gradient Descent Explained (ibm.com)

Gradient descent variants





In stochastic gradient descent (SGD), unlike gradient descent (GD), there's no guarantee that the objective function will decrease with each step. But gradually objective function is decreasing

Also, SGD is more sensitive to step size compared to GD In simple terms, stochastic gradient descent utilizes stochastic gradients

Here, "stochastic" simply refers to some randomness involved in the process

Gradient Descent Explained (ibm.com)

Image is adapted from Lau, S., Gonzalez, J., Nolan, D. (2020) https://www.textbook.ds100.org

Linear Regression:

- For one independent variable (one dimensional, D=1)

$$y(\mathbf{x}) = w_0 + w_1 x_1 + \cdots, + w_D x_D = \mathbf{w}^T \mathbf{x}$$

$$y(\mathbf{x}) = w_0 + w_1 x_1$$

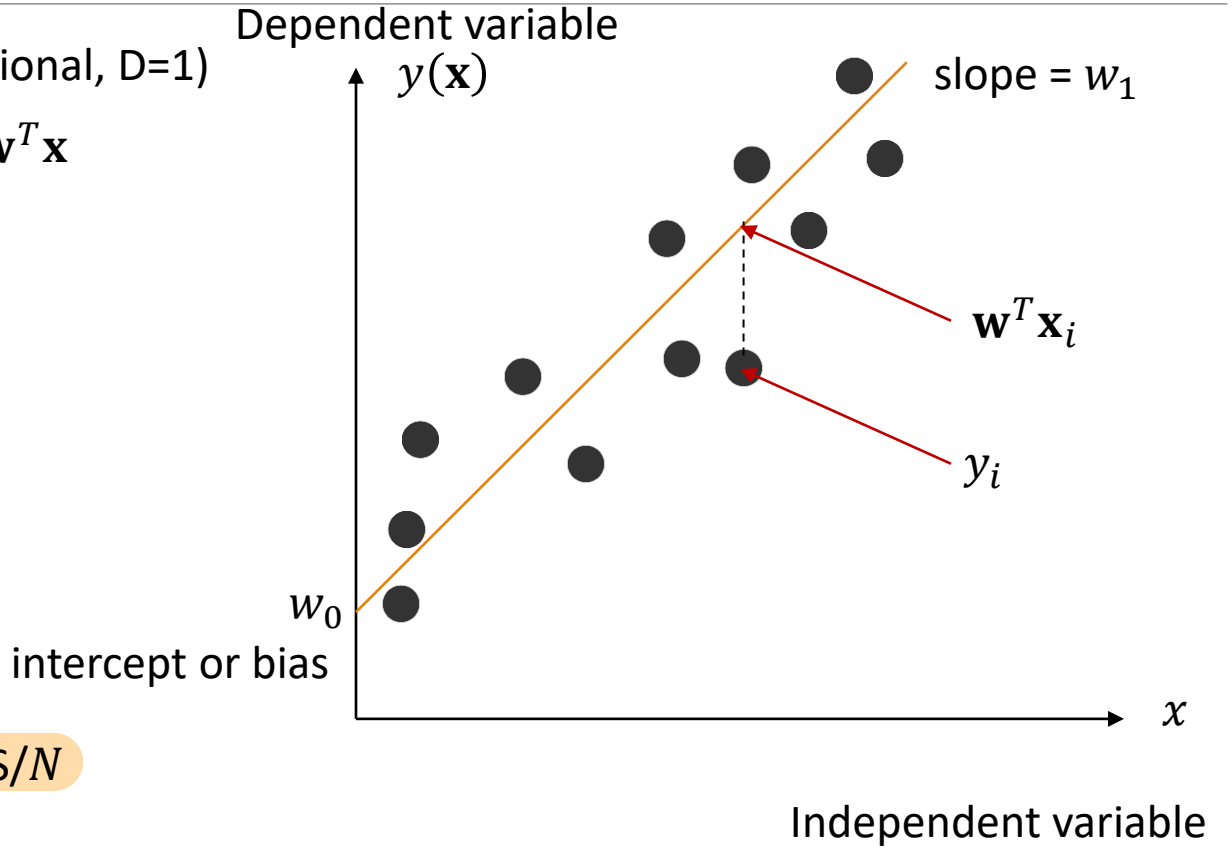
without dividing by N

Residual Sum of Squares (RSS) or
Sum of Squared Error (SSE)

$$\text{RSS}(\mathbf{w}) \triangleq \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)^2$$

Mean Squared Error (MSE) = SSE/N or RSS/N

after dividing by N



Linear Regression: Normal equations

$$\mathrm{RSS}(\mathbf{w}) \triangleq \sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2 = \left(y_1 - [w_0\ w_1 \dots w_D]\begin{bmatrix}1\\x_{1,1}\\.\\.\\.\\x_{D-1,1}\end{bmatrix}\right)^2 + \left(y_2 - [w_0\ w_1 \dots w_D]\begin{bmatrix}1\\x_{1,2}\\.\\.\\.\\x_{D-1,2}\end{bmatrix}\right)^2 + \dots + \left(y_N - [w_0\ w_1 \dots w_D]\begin{bmatrix}1\\x_{1,N}\\.\\.\\.\\x_{D-1,N}\end{bmatrix}\right)^2$$

(Error: First data sample)^2 (Error: Second data sample)^2 (Error: N-th data sample)^2

$$= ||\mathbf{y} - \boldsymbol{X}\boldsymbol{w}||_2^2$$

$$= (\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$\mathbf{y} = \begin{bmatrix}y_1\\.\\y_i\\.\\y_N\end{bmatrix} \qquad \boldsymbol{X} = \begin{bmatrix}1 & x_{1,1} & x_{D,1}\\1 & x_{1,2} & x_{D,2}\\. & . & .\\. & . & .\\. & . & .\\1 & x_{1,N} & x_{D,N}\end{bmatrix}$$

- $\boldsymbol{w}$ are estimated by minimizing the RSE
- How ? Setting the gradient to zero
- Gradient $\mathbf{g}(\mathbf{w}) = 0$

matrixid.pdf (nyu.edu)

Linear Regression: Normal equations

$$L(\mathbf{w}) = (\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$= \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\mathbf{w} - (\mathbf{X}\mathbf{w})^T\mathbf{y} + (\mathbf{X}\mathbf{w})^T\mathbf{X}\mathbf{w}$$

$$= \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\mathbf{w} - \mathbf{w}^T\mathbf{X}^T\mathbf{y} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$$

$$\frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} = (-\mathbf{y}^T\mathbf{X})^T - \mathbf{X}^T\mathbf{y} + \mathbf{X}^T\mathbf{X}\mathbf{w} + (\mathbf{w}^T\mathbf{X}^T\mathbf{X})^T = -2\,\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\mathbf{w}$$

pseudo-inverse

$$\frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} = \mathbf{0} \Rightarrow (\mathbf{X}^T\mathbf{X})\mathbf{w} = \mathbf{X}^T\mathbf{y}$$ normal equations

$$\hat{\mathbf{w}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$ Known as ordinary least squares

derivative of the vector **y** with respect to vector **x**

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \overset{\text{def}}{=} \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_m}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_1}{\partial x_n} & \frac{\partial y_2}{\partial x_n} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

$\mathbf{y}$	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$
$\mathbf{A}\mathbf{x}$	$\mathbf{A}^T$
$\mathbf{x}^T\mathbf{A}$	$\mathbf{A}$
$\mathbf{x}^T\mathbf{x}$	$2\mathbf{x}$
$\mathbf{x}^T\mathbf{A}\mathbf{x}$	$\mathbf{A}\mathbf{x} + \mathbf{A}^T\mathbf{x}$

Here, $(\mathbf{X}^T\mathbf{X})^{-1}$ should be invertible, for that $\mathbf{X}$ needs to be full rank.

matrixcalculus.pdf (mit.edu)

Linear Regression

Example:

X (independent variable)	y
1	2.5
2	4
3	7.3
4	9
5	11
6	14

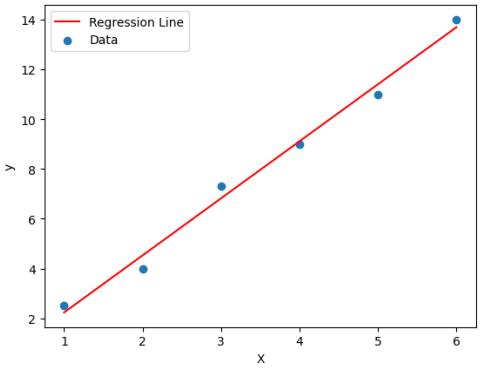
$$\mathbf{y} = \begin{bmatrix} y_1 \\ . \\ . \\ . \\ y_N \end{bmatrix} \qquad \boldsymbol{X} = \begin{bmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ . & . & . \\ . & . & . \\ 1 & x_{1,N} & x_{D,N} \end{bmatrix}$$

$$\hat{\mathbf{w}}_{\mathrm{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

1. Write down $\mathbf{X}$ and $\mathbf{y}$
2. What is $\hat{\mathbf{w}}_{\mathrm{OLS}}$?
3. Predict value of y for x=8.

Here, $(\mathbf{X}^T\mathbf{X})^{-1}$ should be invertible, for that $\mathbf{X}$ needs to be full rank.

Linear Regression



X	y
1	2.5
2	4
3	7.3
4	9
5	11
6	14

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}$$

$$\mathbf{y} = [2.5\ 4\ 7.3\ 9\ 11\ 14]$$

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 47.8 \\ 207.4 \end{bmatrix}$$

$$(\mathbf{X}^T\mathbf{X}) = \begin{bmatrix} 6 & 21 \\ 21 & 91 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \begin{bmatrix} 0.86 & -0.2 \\ -0.2 & 0.057 \end{bmatrix}$$

1. Write down $\mathbf{X}$ and $\mathbf{y}$
2. What is $\hat{\mathbf{w}}_{\mathrm{OLS}}$?
3. Predict value of y for x=8.

$$\hat{\mathbf{w}}_{\mathrm{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$= \begin{bmatrix} 0.86 & -0.2 \\ -0.2 & 0.057 \end{bmatrix}\begin{bmatrix} 47.8 \\ 207.4 \end{bmatrix} = \begin{bmatrix} -0.053 \\ 2.29 \end{bmatrix}$$

$\hat{y}$ = −0.053 + 2.29 x 8 = 18.267

Linear Regression

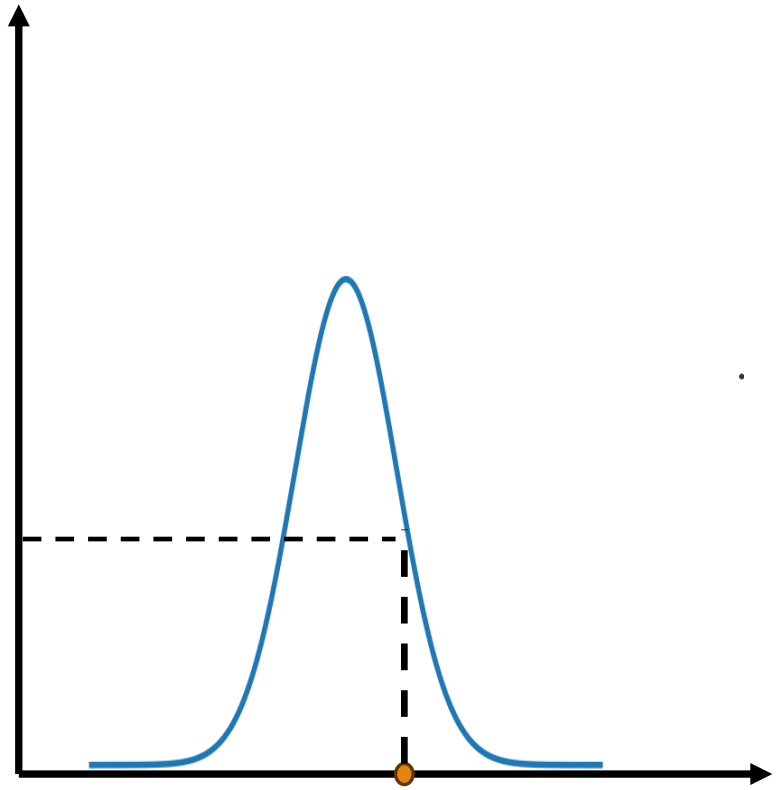
- $y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + \epsilon$ ➔ $\epsilon = y(\mathbf{x}) - \mathbf{w}^T\mathbf{x}$
- Gaussian noise model: assume that error term ϵ follows a Gaussian distribution (zero mean and variance σ^2) $p(\epsilon) = \mathcal{N}(0|\,\sigma^2)$
- $p(y|\mathbf{x}, \mathbf{w}) = \mathcal{N}(y|\mathbf{w}^T\mathbf{x}, \sigma^2)$

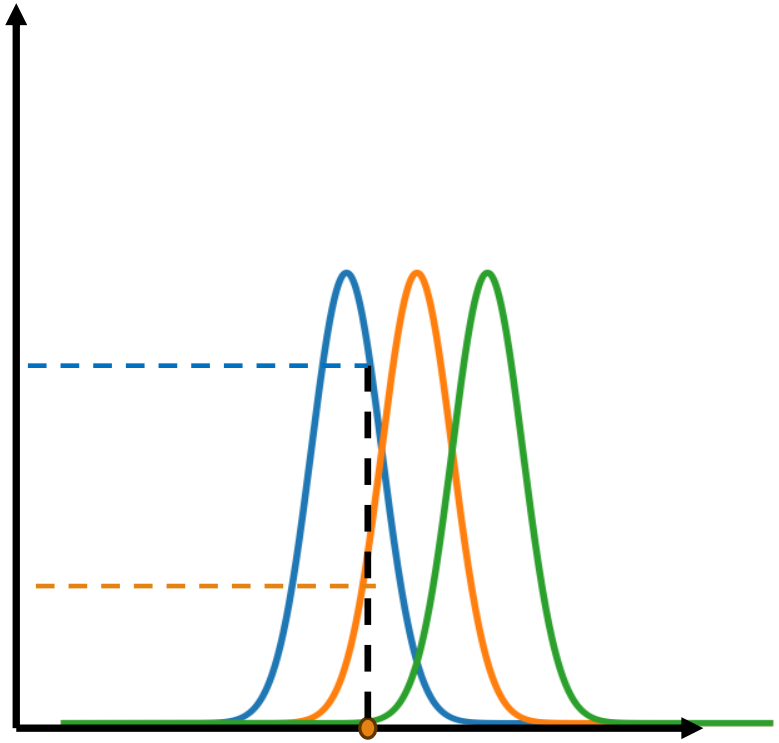


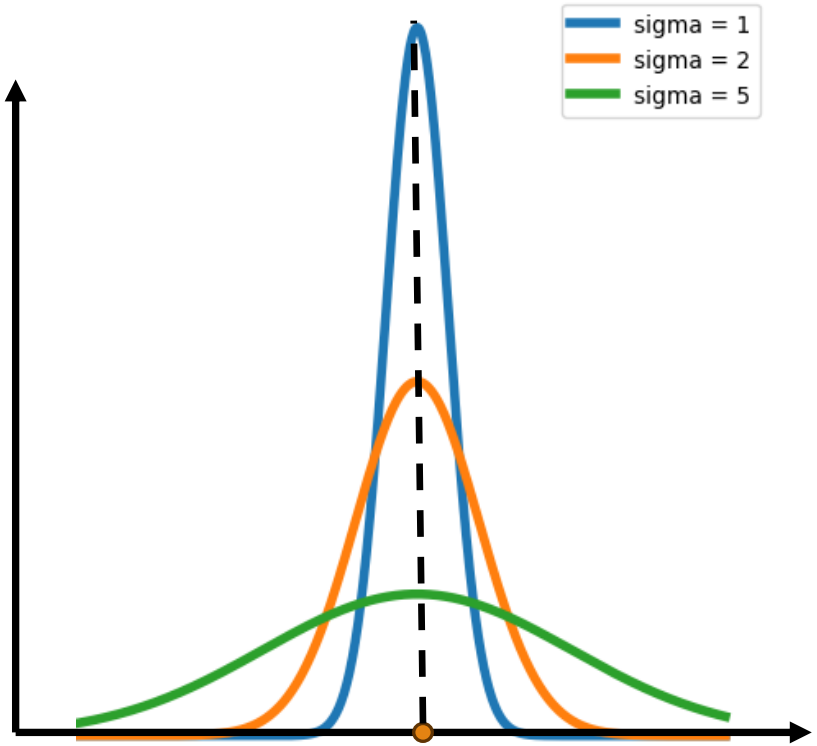
Mean $\mathbf{w}^T\mathbf{x}$ and variance σ^2

conditional probability of the output variable y given the input variable $\mathbf{x}$ and model parameters $\mathbf{w}$

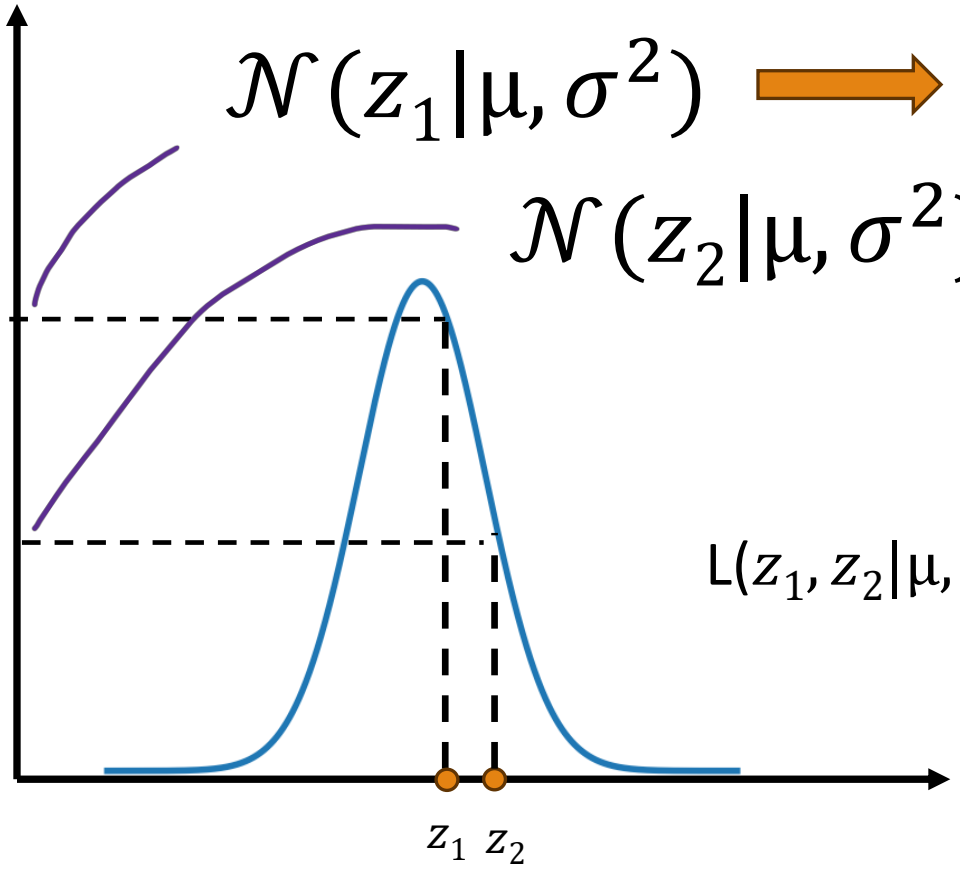
Linear Regression: Likelihood







Linear Regression: Likelihood



$$\frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(z_1-\mu)^2}{2\sigma^2}}$$

$$\frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(z_2-\mu)^2}{2\sigma^2}}$$

$$L(z_1, z_2|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(z_1-\mu)^2}{2\sigma^2}} \quad X \quad \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(z_2-\mu)^2}{2\sigma^2}}$$

$$L(z_1, z_2, \ldots z_N|\mu,\sigma^2) = \prod_{i=1}^{N} \mathcal{N}(z_i|\mu,\sigma^2) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(z_i-\mu)^2}{2\sigma^2}}$$

Linear Regression: Likelihood

$L(y_1, y_2, \ldots y_N | \mathbf{w}^T \mathbf{x}, \sigma^2) = \prod_{i=1}^{N} \mathcal{N}(y_i | \mathbf{w}^T \mathbf{x}_i, \sigma^2)$

Log likelihood = ln $(\prod_{i=1}^{N} \mathcal{N}(y_i | \mathbf{w}^T \mathbf{x}_i, \sigma^2))$ = ln $\left(\prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \mu)^2}{2\sigma^2}} \right)$

$$= N \ln \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \sum_{i=1}^{N} \frac{(y_i - \mu)^2}{2\sigma^2}$$

$$= -\frac{N}{2} \ln (\sqrt{2\pi}) - \frac{N}{2} \ln (\sigma) - \sum_{i=1}^{N} \frac{(y_i - \mu)^2}{2\sigma^2}$$

Linear Regression: Likelihood

$$L(y_1, y_2, \ldots y_N | \mathbf{w}^T\mathbf{x}, \sigma^2) = -\frac{N}{2}\ln(\sqrt{2\pi}) - \frac{N}{2}\ln(\sigma) - \sum_{i=1}^{N}\frac{(y_i-\mu)^2}{2\sigma^2}$$

$\mu = (\mathbf{w}^T\mathbf{x})$ and $\mathbf{x}$ is given too.

In vector-matrix form (with a ½, NLL – Negative Log Likelihood)

$$\mathrm{NLL}(\mathbf{w}) = \frac{1}{2}(\mathbf{y}-\mathbf{X}\mathbf{w})^T(\mathbf{y}-\mathbf{X}\mathbf{w})$$

- In this case, NLL $(\mathbf{w})$ is equivalent to **residual sum of squares (RSS)***
- How to find $\mathbf{w}$ (assuming σ is fixed),Setting the gradient to zero
- Gradient $\mathbf{g}(\mathbf{w}) = (\mathbf{X}^T\mathbf{X})\mathbf{w} - \mathbf{X}^T\mathbf{y}$
- $\mathbf{g}(\mathbf{w}) = 0 \Rightarrow (\mathbf{X}^T\mathbf{X})\mathbf{w} = \mathbf{X}^T\mathbf{y}$
- $\widehat{\mathbf{w}}_{\mathrm{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

Maximum likelihood estimator (MLE)**

*with certain constant factors in the expression are disregarded

**Minimizing the negative log-likelihood is equivalent to maximizing the likelihood

Linear Regression

$$\hat{\mathbf{w}}_{\mathrm{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{X}^{\dagger}\,\mathbf{y}$$

pseudo inverse of the (non-square) matrix $\mathbf{X}$

- Condition for unique solution
  - $\mathbf{X}$ is a full rank matrix* (columns of $\mathbf{X}$ are linearly independent)
  - Full rank matrix ➔ No row or column can be expressed as a linear combination of the other rows or columns, respectively

$$\boldsymbol{X} = \begin{bmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ . & . & . \\ . & . & . \\ . & . & . \\ 1 & x_{1,N} & x_{D,N} \end{bmatrix}$$

full rank matrix, where rank is equal to the minimum of the number of its rows and columns

Linear Regression: Likelihood

- Deriving the MLE for σ^2

$$L(y_1, y_2, \ldots y_N | \hat{\mathbf{w}}^T \mathbf{x}, \sigma^2) = -\frac{N}{2}\ln(\sqrt{2\pi}) - \frac{N}{2}\ln(\sigma) - \sum_{i=1}^{N} \frac{(y_i - \hat{\mathbf{w}}^T \mathbf{x}_i)^2}{2\sigma^2}$$

$$\sigma^2 = \arg\min_{\sigma^2} \mathrm{NLL}(\hat{\mathbf{w}}, \sigma^2) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{\mathbf{w}}^T \mathbf{x}_i)^2$$

Linear Regression: Solving w_0 separately

- $\mathcal{N}(y|w_0 + w_1 x_1 + \cdots, + w_D x_D, \sigma^2)$

- $L(y_1, y_2, \ldots y_N|\mathbf{w}^T\mathbf{x}, \sigma^2) = -\frac{N}{2}\ln(\sqrt{2\pi}) - \frac{N}{2}\ln(\sigma) - \sum_{i=1}^{N}\frac{(y_i - \mathbf{w}^T\mathbf{x}_i)^2}{2\sigma^2}$

$$= -\frac{N}{2}\ln(\sqrt{2\pi}) - \frac{N}{2}\ln(\sigma) - \sum_{i=1}^{N}\frac{(y_i - (w_0 + w_1 x_{1i} + \cdots, + w_D x_{Di}))^2}{2\sigma^2}$$

$$= -\frac{N}{2}\ln(\sqrt{2\pi}) - \frac{N}{2}\ln(\sigma) - \sum_{i=1}^{N}\frac{(y_i - (w_0 + \tilde{\mathbf{w}}^T\tilde{\boldsymbol{x}}_i))^2}{2\sigma^2}$$

- Solving w_0 (Intercept or bias) and feature coefficients are separately

$$\frac{\partial L}{\partial w_0} = 0 \qquad w_0 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \tilde{\mathbf{w}}^T\tilde{\boldsymbol{x}}_i) = \bar{y} - \frac{1}{N}\sum_{i=1}^{N}(\tilde{\mathbf{w}}^T\tilde{\boldsymbol{x}}_i) = \bar{y} - \tilde{\mathbf{w}}^T\bar{x}$$

Average of the model outcome

Average of the target values

Linear Regression

- $\tilde{\mathbf{w}}^{T} = [w_1, .., w_D]$
- Solving feature coefficients are separately, estimated $\tilde{\mathbf{w}}$ is given by
- $\hat{\tilde{\mathbf{w}}} = \left[\sum_{i=1}^{N}(\tilde{\boldsymbol{x}}_i - \bar{\boldsymbol{x}})(\tilde{\boldsymbol{x}}_i - \bar{\boldsymbol{x}})\,T\right]^{-1}\left[\sum_{i=1}^{N}(y_i - \bar{y})(\tilde{\boldsymbol{x}}_i - \bar{\boldsymbol{x}})\right] = (\mathbf{X}_C{}^{T}\mathbf{X}_C)^{-1}\mathbf{X}_C{}^{T}\mathbf{y}_C$

Centered output vector

Centered input matrix

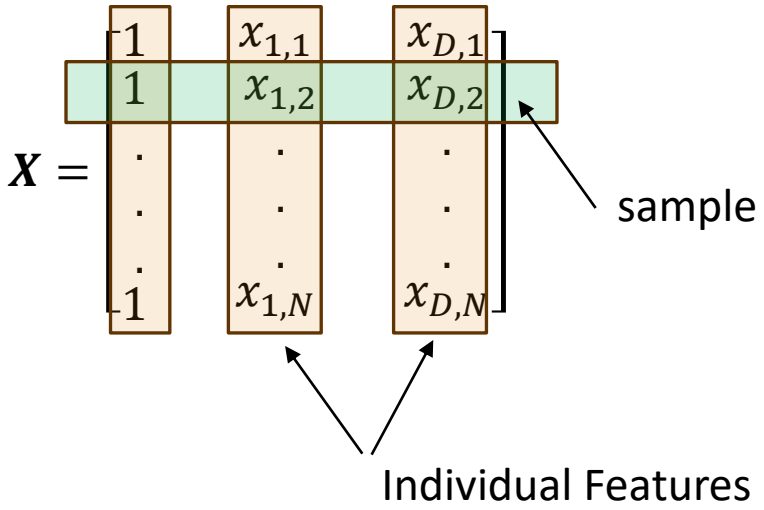
- For 1D $y = w_0 + w_1 x$

$\widehat{w_1} = \sum_{i=1}^{N}(yi - \bar{y})(x - \bar{x}) / \sum_{i=1}^{N}(x - \bar{x})^2$ = Cov [X; Y] / Cov [X; X] =C_{xy} /C_{xx}

$\widehat{w_0} = \bar{y} - \widehat{w_1}\bar{x}$

Procedure: Estimating $\tilde{\mathbf{w}}$ by centered data then estimating the w_0 by $\bar{y} - \tilde{\mathbf{w}}^{T}\bar{x}$

Linear Regression: Geometric interpretation



$$\boldsymbol{X} = \begin{bmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ . & . & . \\ . & . & . \\ . & . & . \\ 1 & x_{1,N} & x_{D,N} \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ . \\ . \\ . \\ y_N \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ . \\ . \\ . \\ w_D \end{bmatrix}$$

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{1,i} \\ . \\ . \\ . \\ x_{D,i} \end{bmatrix}$$

$$\widehat{\mathbf{w}}_{\mathrm{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\hat{\mathbf{y}} = \mathbf{X}\ \widehat{\mathbf{w}}_{\mathrm{OLS}} = \mathbf{X}\,(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$$\mathbf{Proj}\,(\mathbf{X}) = \mathbf{X}\,(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

$$\hat{\mathbf{y}} = \mathbf{Proj}\,(\mathbf{X})\,\mathbf{y}$$

The orthogonal projection of vector $\mathbf{y}$ onto the column space of matrix $\mathbf{X}$

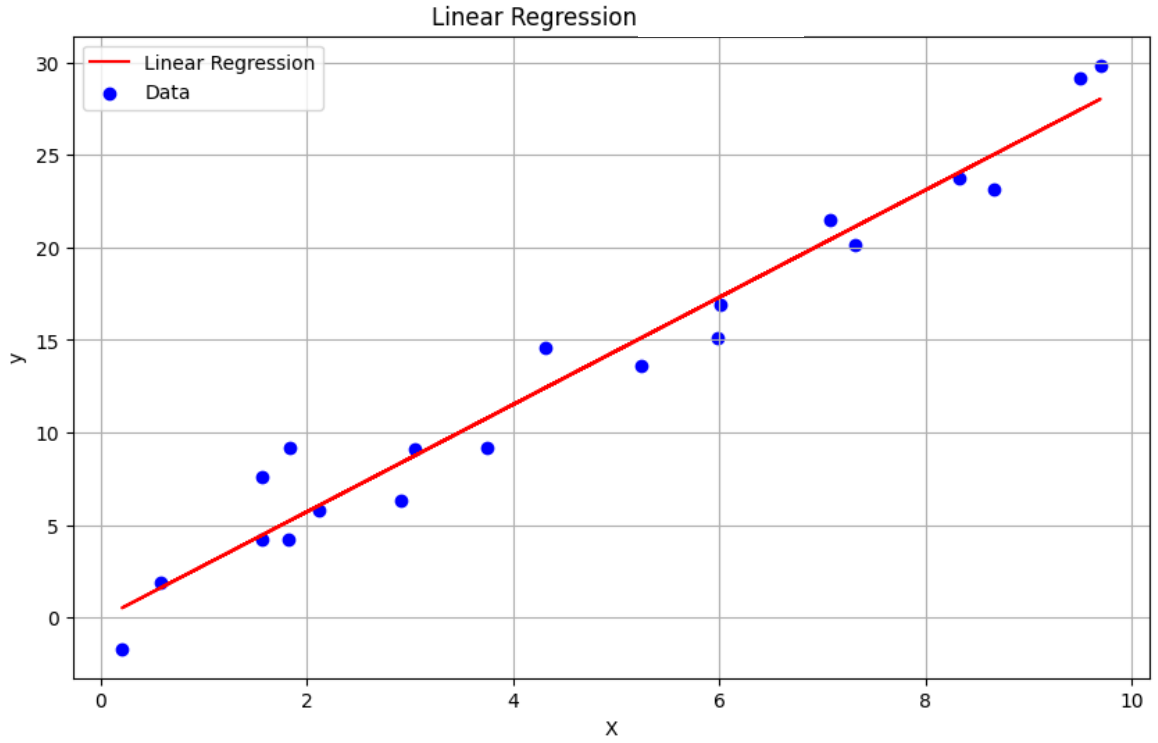
Linear Regression: Implementation

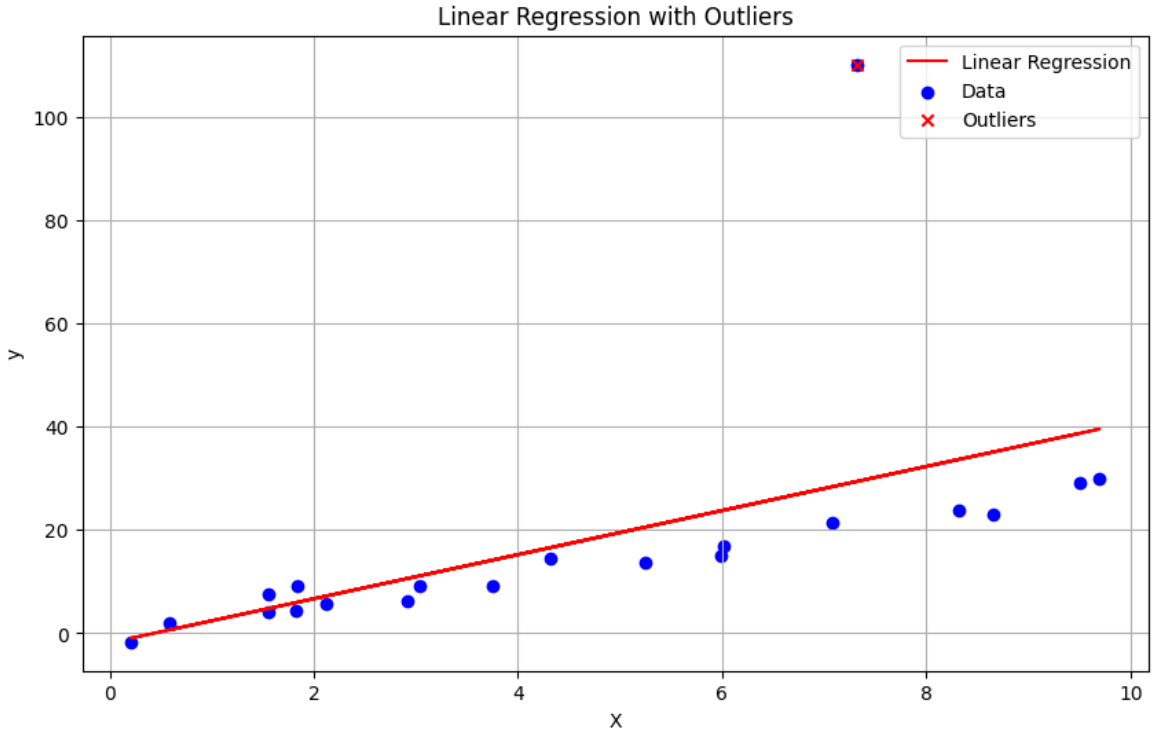
$$\hat{\mathbf{w}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

may be ill conditioned or singular ➔ cause issues in numerical computations

- SVD based pseudo inverse calculation (see sklearn implementation)

Linear Regression: Example





how to calculate in python?

$y \to (0, 100)$ RSS = 5
$y \to (0, 1)$ RSS = 0.1
$y \to (0, 10000)$ RSS = 20

} cannot compare due to different ranges of y
To compare use R^2 statistic

R^2 Statistic

By dividing by total variance of y we normalize the RSS such that it is independent of the range of y

"Amount of variability that is **left unexplained** after performing the regression (amount of unexplained)"

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x})^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} = 1 - \frac{\text{RSS}}{\sum_{i=1}^{N}(y_i - \bar{y})^2} = 1 - \frac{\text{RSS}}{\text{TSS}} \in (0, 1)$$

variance of all y values

(mean of all y values

"Total variance in the y"

TSS – RSS "measures the amount of variability in the response that is explained (or removed) by performing the regression" [1].

"R^2 measures proportion of variability in Y that can be explained using X" ("R^2 is close to 1 mean that the large proportion of the variability in the response has been explained by the regression"[1]. (RSS is low, LR is more accurate)

[1] Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, An Introduction to Statistical Learning with Applications in R, 2013.

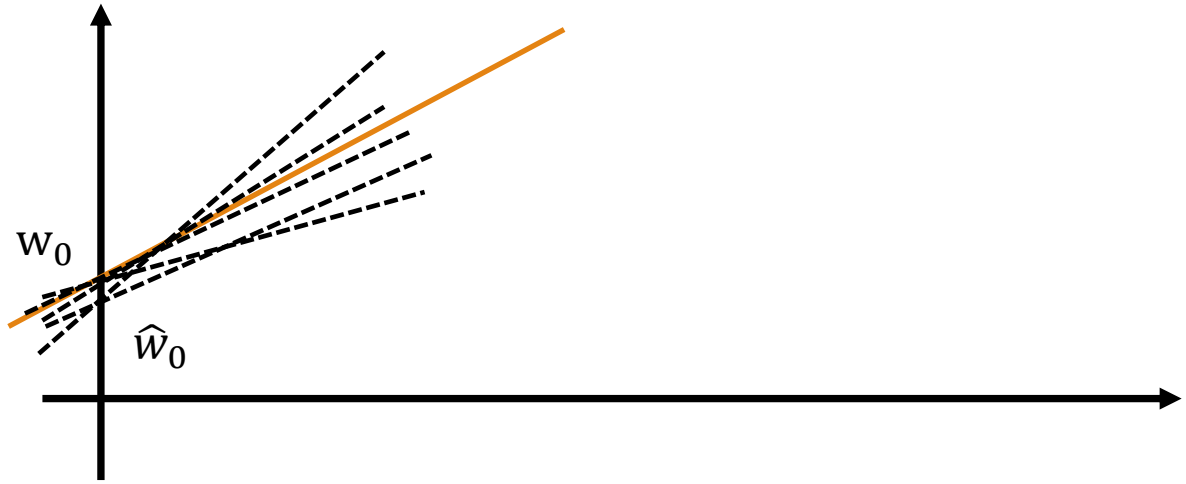
Standard errors

- True relationship $y = w_0 + w_1 x$
- How close our estimate (w_0, w_1) to true values $(\hat{w}_0, \hat{w}_1)$
- Standard errors

$$\mathrm{SE}^2(\hat{w}_0) = \sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^{N} (x_i - \bar{x})^2} \right]$$

$$\mathrm{SE}^2(\hat{w}_1) = \frac{\sigma^2}{\sum_{i=1}^{N} (x_i - \bar{x})^2}$$

$\sigma_{w_1}^2$



σ^2 can be estimated from data using RSE

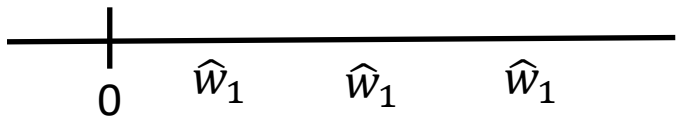
Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, An Introduction to Statistical Learning with Applications in R, 2013.

Is there a true relationship between y, w_0 and w_1

t–statistic

- Standard errors
- Is there a true relationship?
- Hypothesis testing ← Null hypothesis
  - H_0: No relationship between y and w_1 ➔ $w_1 = 0$
  - H_1: There is a relationship between $y$and w_1 ➔ $w_1 \neq 0$

$$\mathrm{SE}^2(\hat{w}_0) = \sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}\right] \qquad \mathrm{SE}^2(\hat{w}_1) = \frac{\sigma^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$



t–statistic

$$t = \frac{\hat{w}_1 - 0}{\mathrm{SE}^2(\hat{w}_1)}$$

number of standard deviations that $\hat{w}_1$ is away from 0

Need to determine $\hat{w}_1$

1. Sufficiently far from zero ➔ $\hat{w}_1 \neq 0$ *or*
2. Sufficiently *close to zero* $\hat{w}_1$ =0

$\hat{w}_1$ =0.1 Sufficiently close to zero? Far away from zero?
Depends on $\mathrm{SE}^2(\hat{w}_1)$

True $w_1 = 5, \mathrm{SE}^2(\hat{w}_1) = 10$ *is significant*
True =1000, $\mathrm{SE}^2(\hat{w}_1)$ =10 is not significant

Confidence intervals

- Standard errors can be used to calculate confidence level.
- 95% Confidence Interval: Range of values containing true parameter value with 95% probability
- The range is calculated from training data.
- For linear regression

For w_0: $\widehat{w}_0 \pm \ \ 2\ \mathrm{SE}^2(\widehat{w}_0)$

For w_1: $\widehat{w}_1 \pm \ \ 2\ \mathrm{SE}^2(\widehat{w}_1)$

p -value

t−statistic

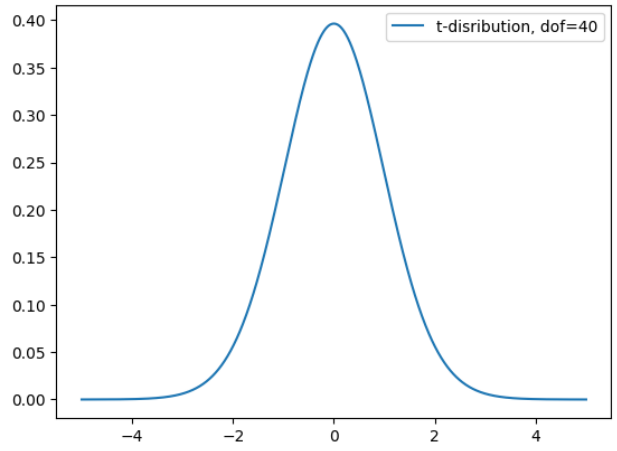
$$t = \frac{\widehat{w}_1 - 0}{\mathrm{SE}^2(\widehat{w}_1)}$$

number of standard deviations that $\widehat{w}_1$ is away from 0

t-distribution with N - 2 degrees of freedom if there is no relationship between x and y

p-value "the probability of observing any values equal to $|t|$ or larger, assuming true $w_1 = 0$. (no association between the predictor and the response variable)"

small p -value reject null hypothesis H_0: No relationship between y and $w_1 (w_1 = 0)$
Typical values 1% or 5%.



Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, An Introduction to Statistical Learning with Applications in R, 2013.

Linear Regression: Example

California housing dataset

#Attribute Information: (independent variables)

- MedInc median income in block group

- HouseAge median house age in block group

- AveRooms average number of rooms per household

- AveBedrms average number of bedrooms per household

- Population block group population

- AveOccup average number of household members

- Latitude block group latitude

- Longitude block group longitude

Dependent variable is median house value (y)

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24
5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25
3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25

y
4.526
3.585
3.521
3.413

$$\boldsymbol{X} = \begin{bmatrix} 1 & x_{1,1} & x_{D,1} \\ 1 & x_{1,2} & x_{D,2} \\ . & . & . \\ . & . & . \\ . & . & . \\ 1 & x_{1,N} & x_{D,N} \end{bmatrix} \qquad \boldsymbol{X} = \begin{bmatrix} 1 & \mathbf{8.3252} & \mathbf{-122.23} \\ 1 & \mathbf{8.3014} & \mathbf{-122.22} \\ . & . & . \\ . & . & . \\ . & . & . \\ 1 & \mathbf{3.8462} & \mathbf{-122.25} \end{bmatrix}$$

https://colab.research.google.com/drive/1tIA-BwokQyrY4zG2zjfdogakAsN-cbBl?usp=sharing

Linear Regression: Example

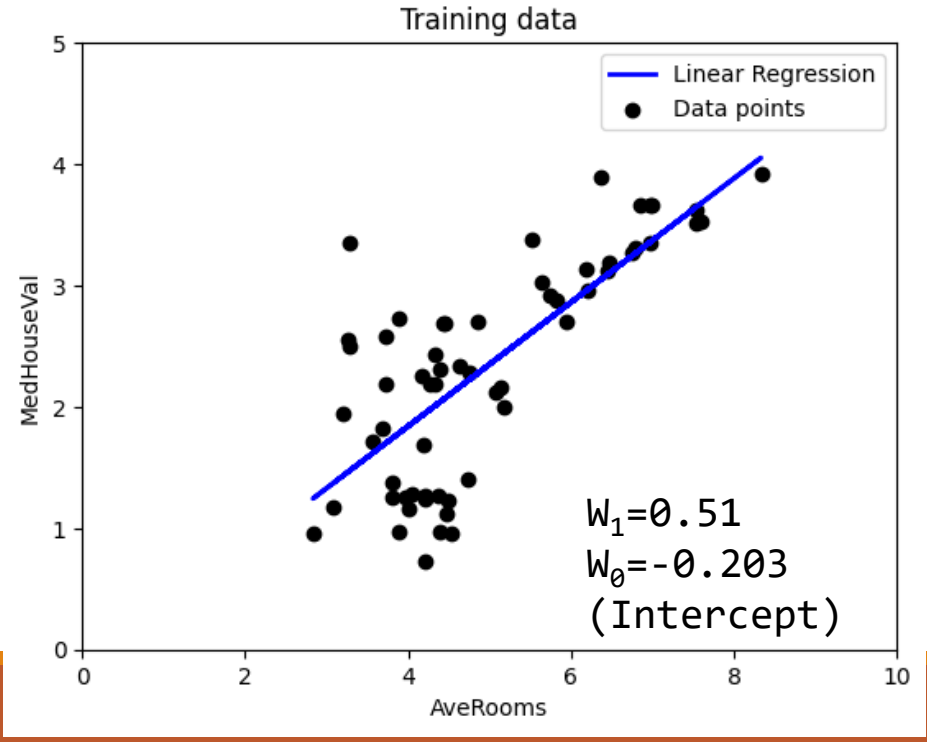
https://colab.research.google.com/drive/1tIA-BwokQyrY4zG2zjfdogakAsN-cbBl?usp=sharing

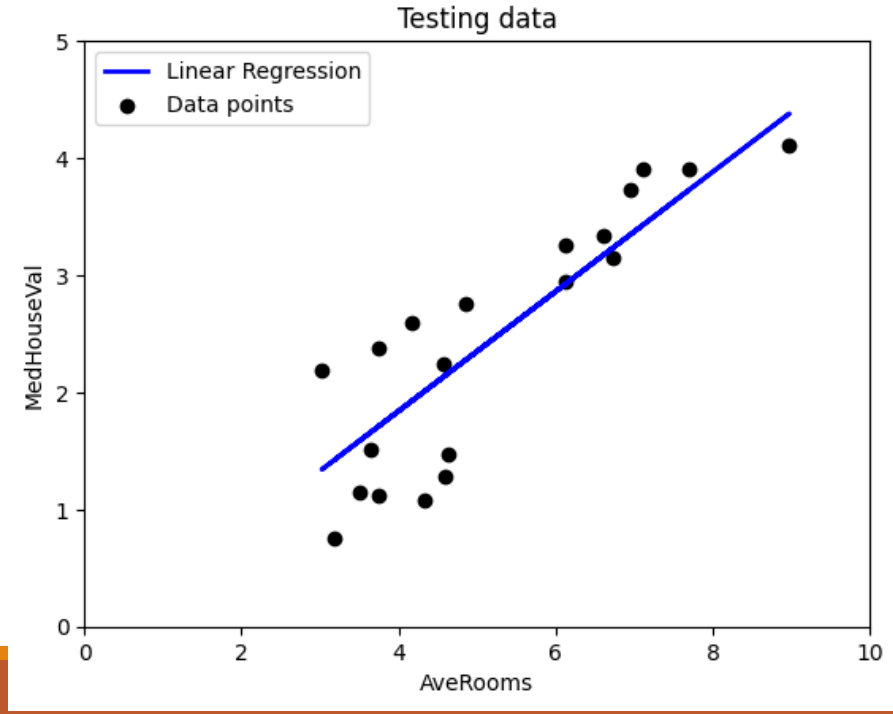
Linear Regression: Example

California housing dataset (100 data samples were taken)

Independent variables- AveRooms

Dependent variable is median house value





Linear Regression: Example

Independent variables- AveRooms, Dependent variable is median house value

- RSS 21.89 RSE 0.61 TSS 49.85 $R^2 = 0.56$
- standard errors for intercept and w_1: 0.089 0.0034
- t-statistic for intercept and w_1: -0.68 8.75
- pvale for intercept and w_1 : 0.25 $1.69e^{-12}$

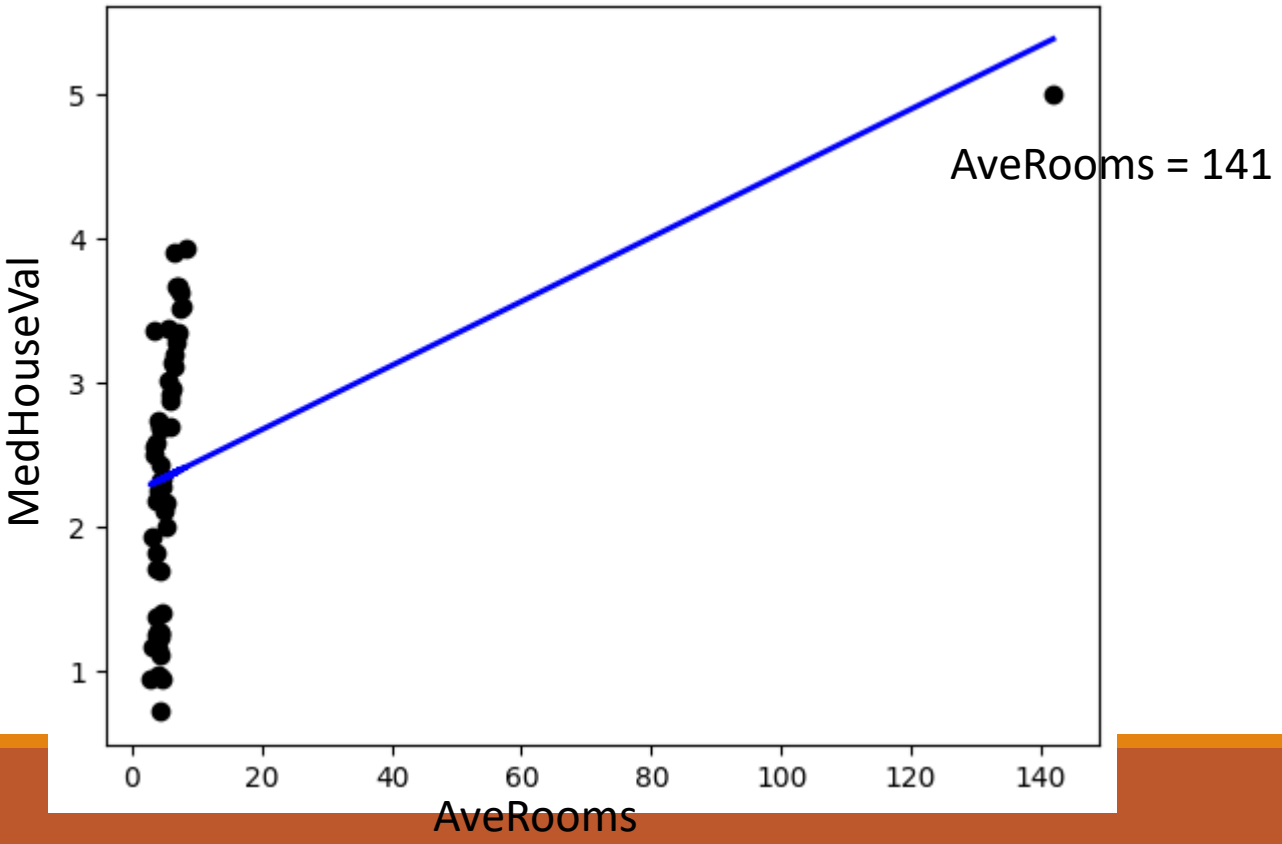
P-value for intercept = 0.25
not statistically significant at conventional significance levels (such as 0.05 or 0.01)
There's no strong evidence to suggest that the constant term in the model significantly affects the response variable. This could imply that the relationship between the predictor and the response starts from or passes through the origin (0,0)

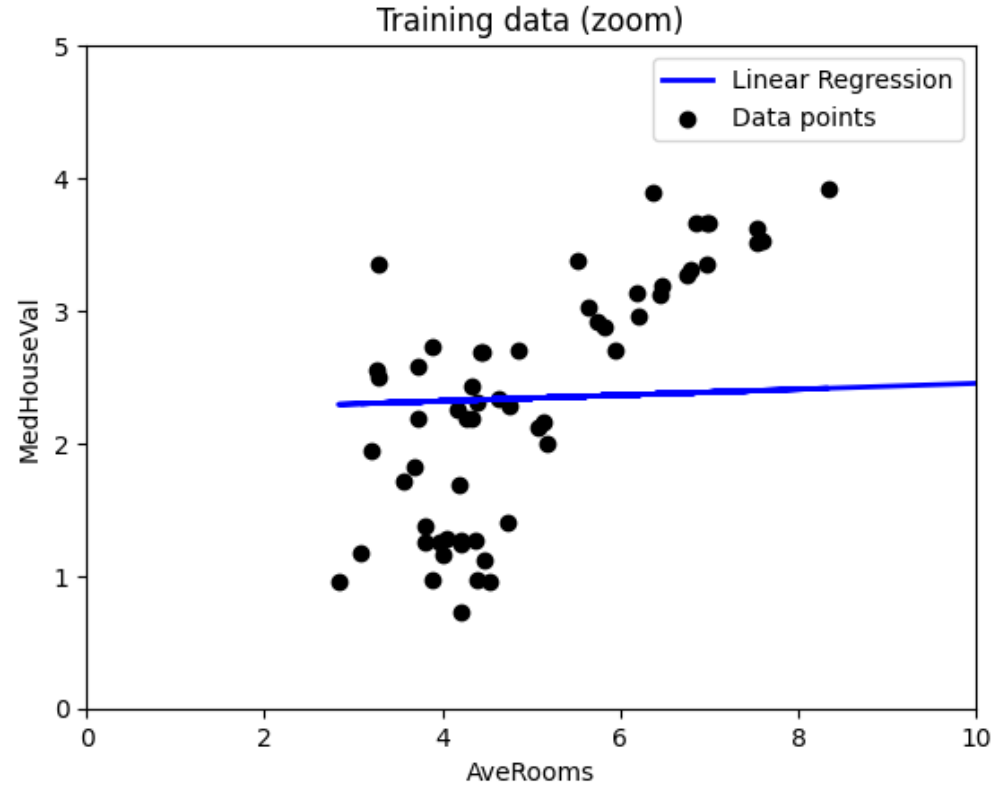
P-value for w_1 = 1.69e-12
coefficient w_1 is likely to be statistically significant and suggests that there is a meaningful linear relationship between the predictor variable (w_1) and the response variable

Linear Regression: Example

Independent variables- AveRooms, Dependent variable is median house value

Impact of outliers





Linear Regression: Example

Independent variables: MedInc HouseAge AveRooms AveBedrms Population AveOccup Latitude Longitude

Dependent variable: MedHouseVal

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
Coefficients:	$4.52e^{-1}$	$9.44e^{-3}$	$-1.30e^{-1}$	$8.24e^{-1}$	$-5.84e^{-6}$	$-7.62e^{-3}$	$-4.06e^{-1}$	$-4.187e^{-1}$

	coef	std err	t	P>|t|	[0.025	0.975]
const	-35.6684	0.849	-42.006	0.000	-37.333	-34.004
x1	0.4527	0.006	81.204	0.000	0.442	0.464
x2	0.0094	0.001	16.493	0.000	0.008	0.011
x3	-0.1302	0.008	-16.185	0.000	-0.146	-0.114
x4	0.8249	0.041	20.252	0.000	0.745	0.905
x5	-5.847e-06	6.15e-06	-0.951	0.341	-1.79e-05	6.2e-06
x6	-0.0076	0.001	-6.442	0.000	-0.010	-0.005
x7	-0.4063	0.009	-43.858	0.000	-0.424	-0.388
x8	-0.4184	0.010	-43.176	0.000	-0.437	-0.399

R-squared: 0.614

Mean squared error: 0.53

A small p-value (usually < 0.05) suggests that the feature is likely to have a significant impact on the target variable. A large p-value suggests that the feature might not have a statistically significant impact on the target variable.

* Minimize the impact of outliers

restriction on size of weight parameters

Ridge Regression

$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2 + \lambda\|\mathbf{w}\|_2^2$$

L2 regularization penalty (shrinkage penalty)

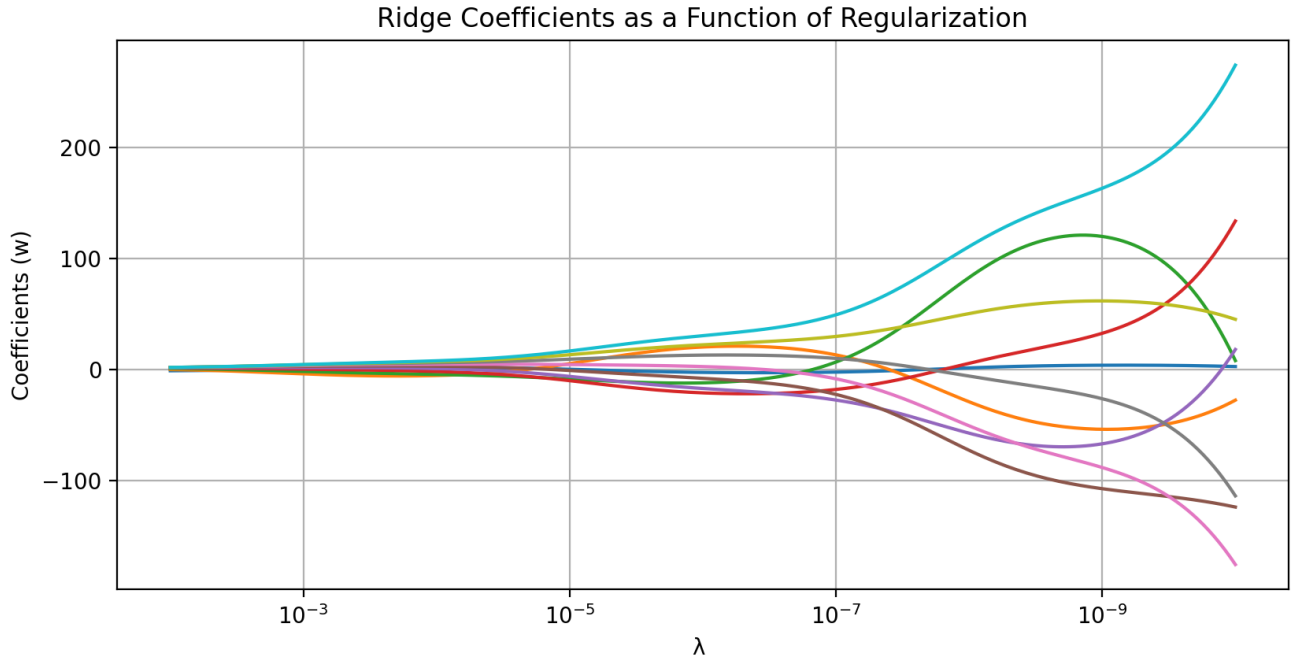
- $\|\mathbf{w}\|_2^2 = \sum_j w_j^2 = \mathbf{w}^T\mathbf{w}$ = the squared two-norm.
- $\lambda \geq 0$, is a complexity penalty (hyper parameter, need to tune).

- Ridge regression also known as Tikhonov regularization, penalized least squares L_2 regularization, or weight decay.
- Address multicollinearity (high correlation between predictor (independent) variables)
- Avoid overfitting
- Regularization discourages the model from assigning overly large coefficients to predictor variables

$$\hat{\mathbf{w}}_{\text{ridge}} = (\lambda\mathbf{I}_D + \mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Ridge Regression: Impact of λ

- $J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2 + \lambda\|\mathbf{w}\|_2^2$
- When λ is very large (coefficinets $\mathbf{w} \to 0$).
- When λ tends toward zero, Ordinary least square ➔ coefficients exhibit big oscillations.
- Need to tune λ to balance



Plot Ridge coefficients as a function of the regularization

Polynomial Regression

- Model: Relationship between the independent variable(s) and the dependent variable is modeled as an nth-degree polynomial*.
- $y = \mathbf{w}^T \phi(\mathbf{x})$ and for single variable $\phi(\mathbf{x}) = [1, x, x^2, \dots, x^d]$
- For two feature $d = 2$, $\phi(\mathbf{x}) = [1, x_1, x_2, x_1^2, x_1 x_2, {x_2}^2]$
- Example $y = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_1 x_2 + w_5 x_2^2$

  $y = w_0 + w_1 z_1 + w_2 z_2 + w_3 z_3 + w_4 z_4 + w_5 z_5$
- Cost function

$$J(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \mathbf{w}^T \phi(\mathbf{x}))^2$$

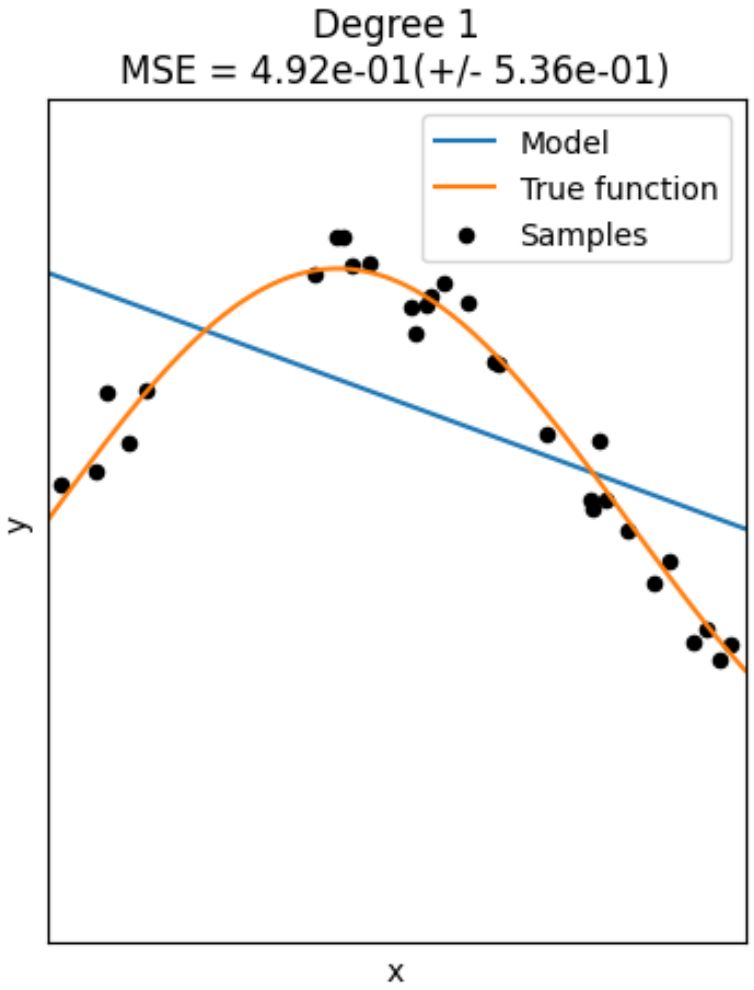
- Polynomial **Ridge** Regression

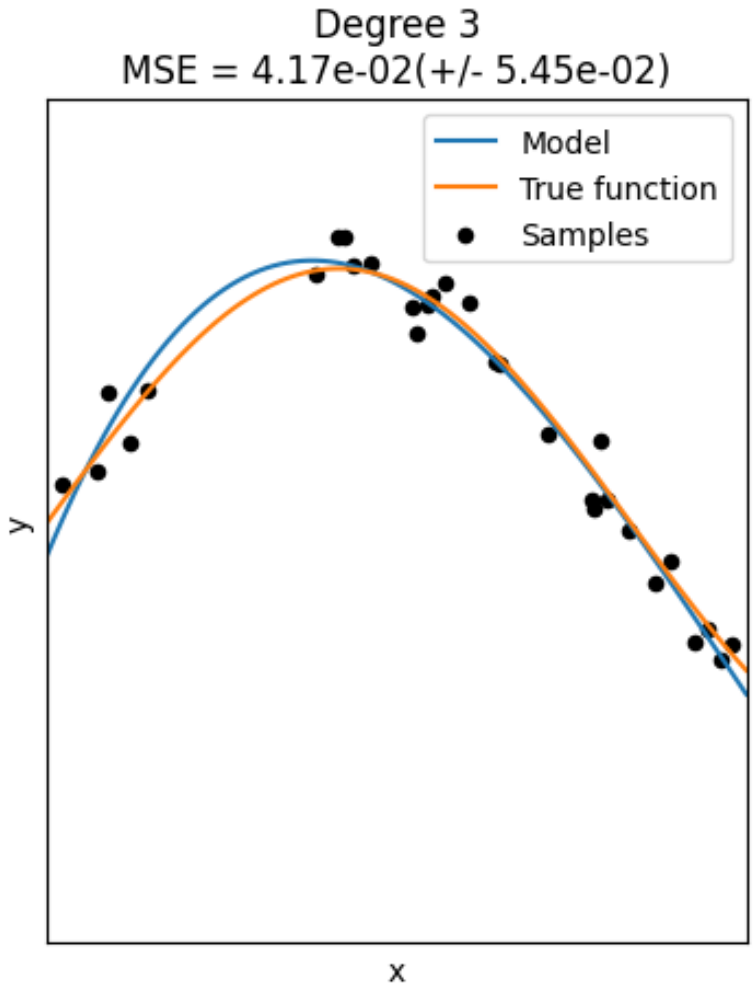
$$J(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \mathbf{w}^T \phi(\mathbf{x}))^2 + \lambda \|\mathbf{w}\|_2^2$$

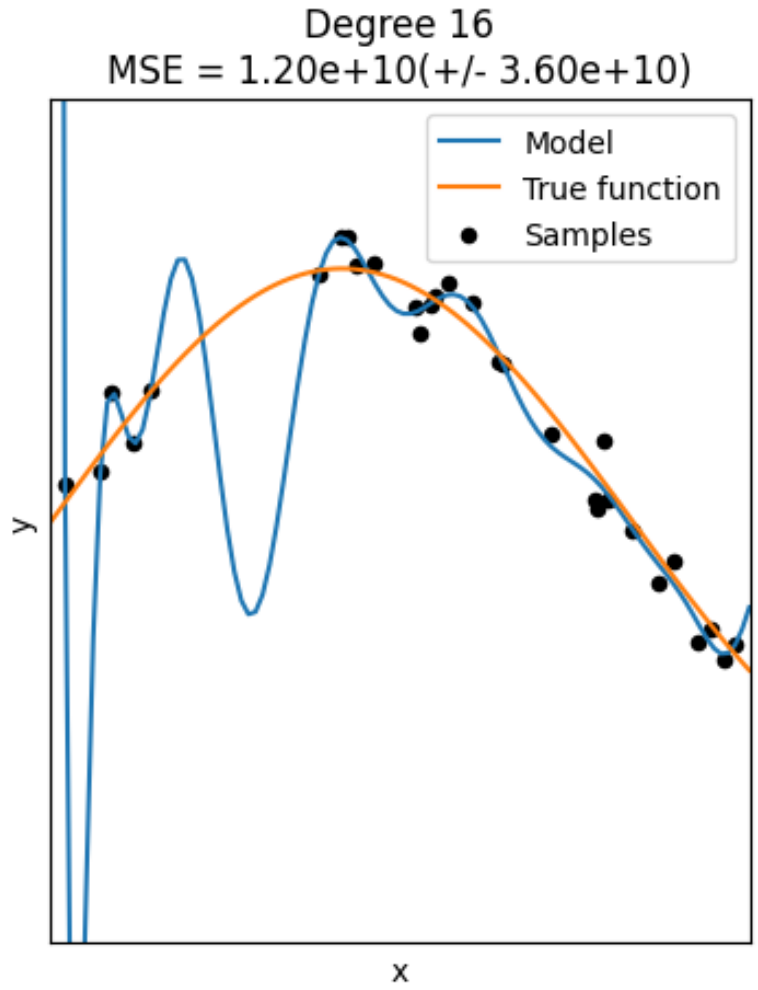
* Still a linear model with respect to weights. Nonlinear model when x_1^w_1

Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, 2012.

Polynomial Regression



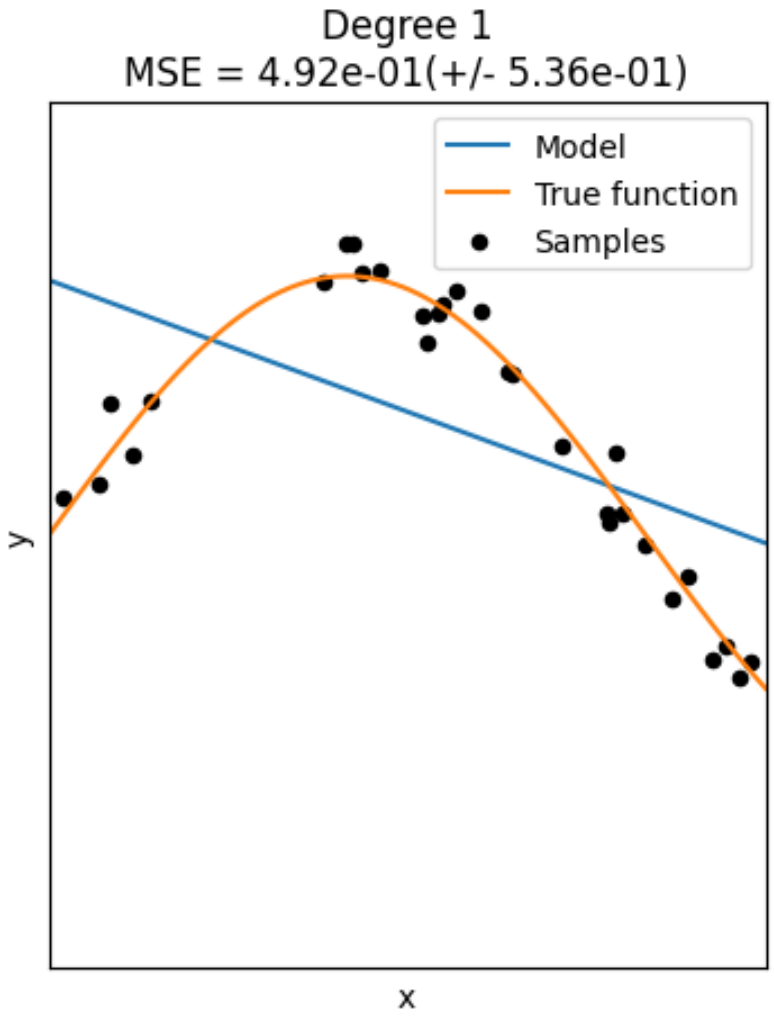


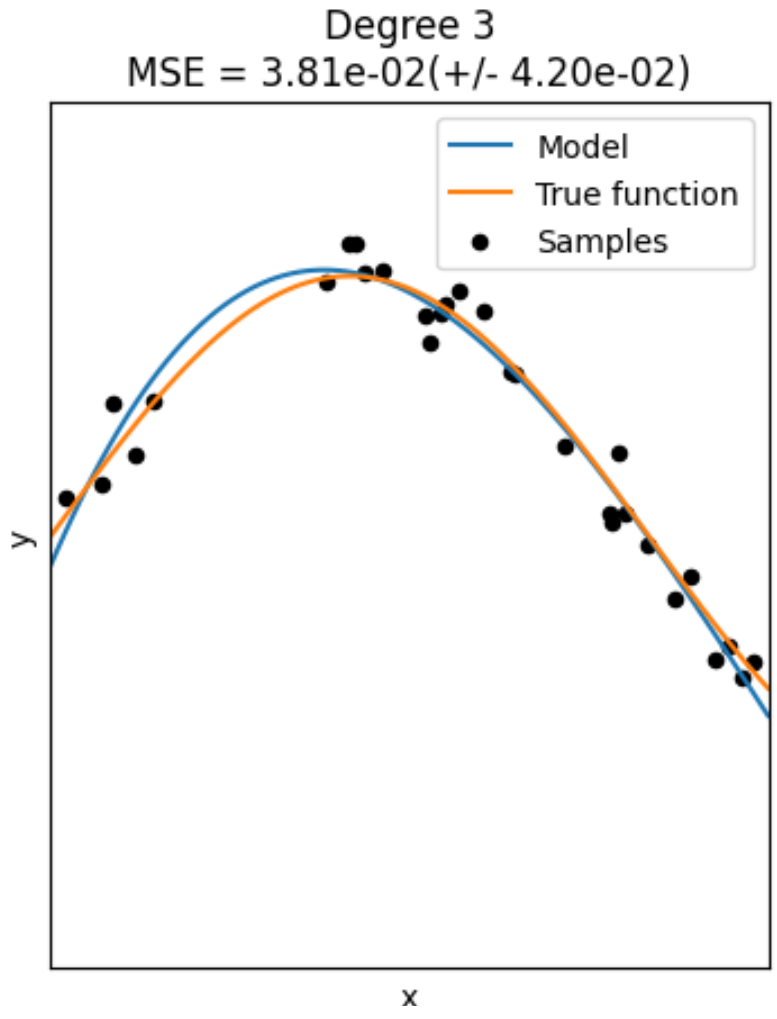


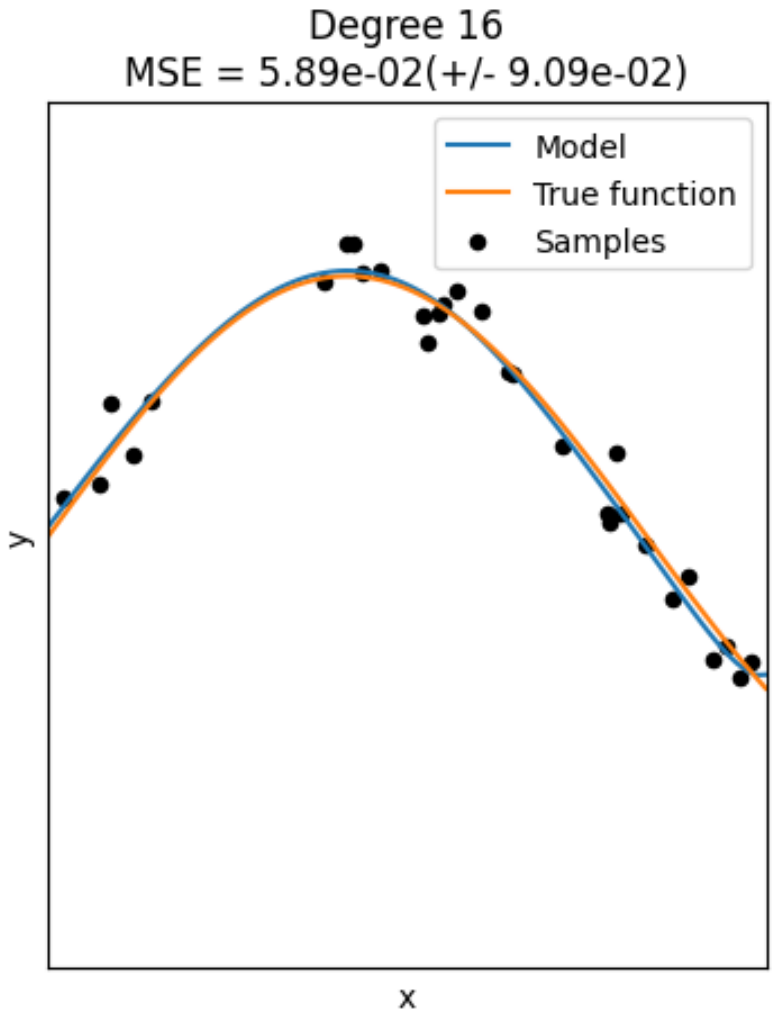
Underfitting vs. Overfitting — scikit-learn 1.3.0 documentation

$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\phi(\mathbf{x}))^2$$

Polynomial Ridge Regression







$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\phi(\mathbf{x}))^2 + \lambda\|\mathbf{w}\|_2^2$$

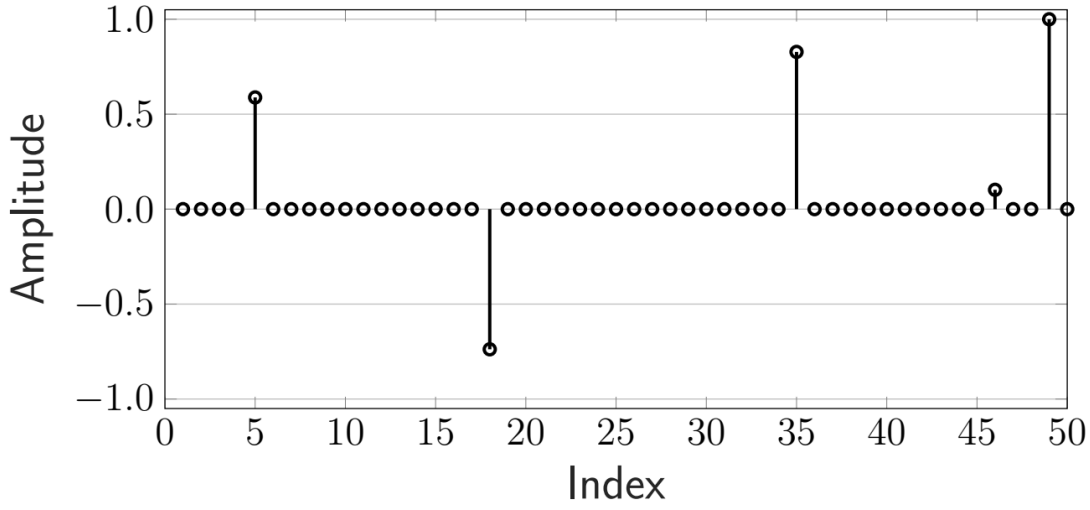
Lasso

- Least Absolute Shrinkage and Selection Operator
- Cost function

$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x})^2 + \lambda\|\mathbf{w}\|_1$$

L1 regularization penalty

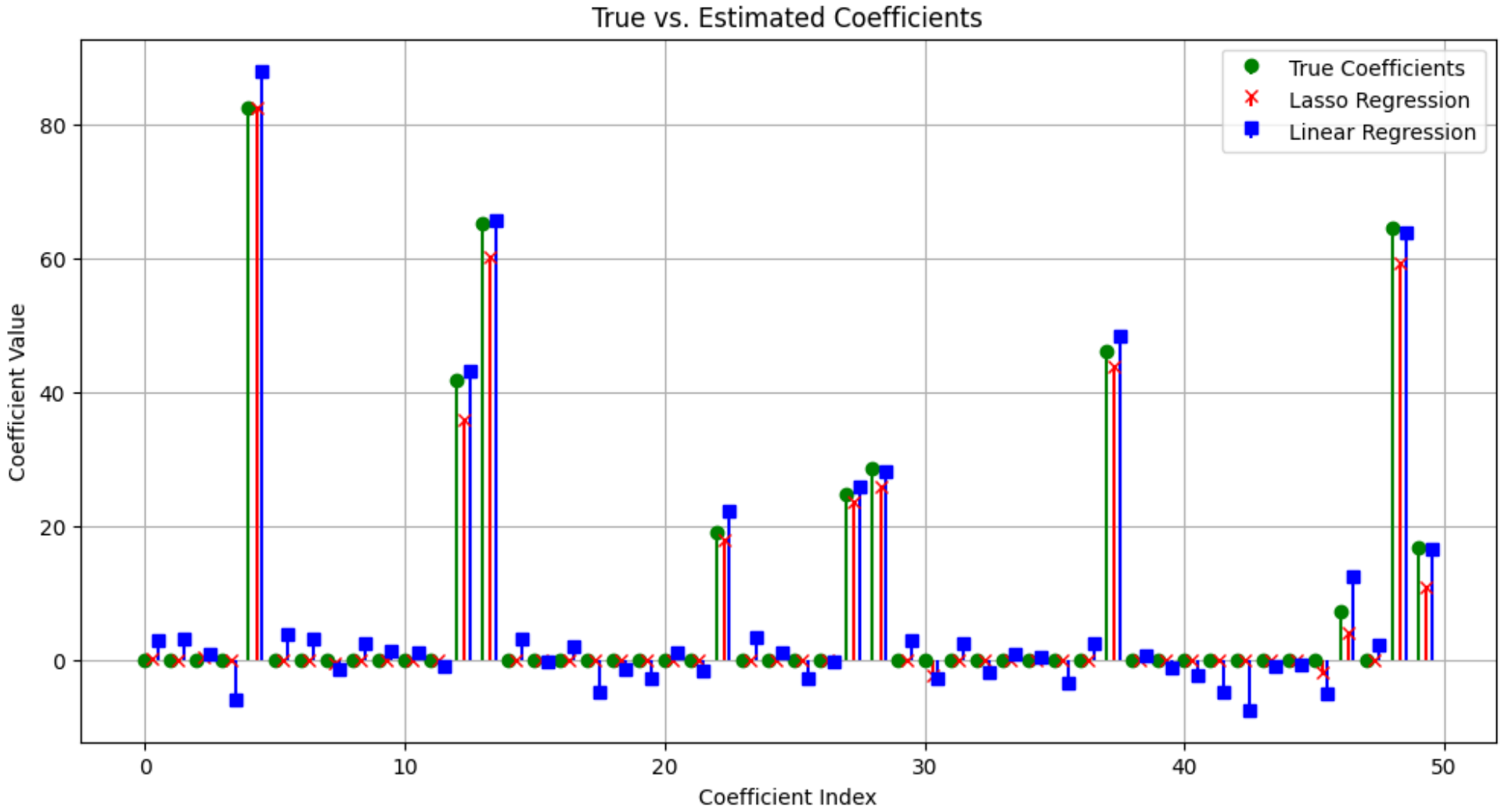
- Select a subset of important features while reducing the coefficients of less important features to nearly zero (Encourages sparsity)
- Useful for feature selection or dimensionality reduction
- Lasso plays important role in compressed sensing



Introduction to Sparsity in Signal Processing (nyu.edu)

Compressive sensing: tomography reconstruction with L1 prior (Lasso) — scikit-learn 1.3.0 documentation

Lasso

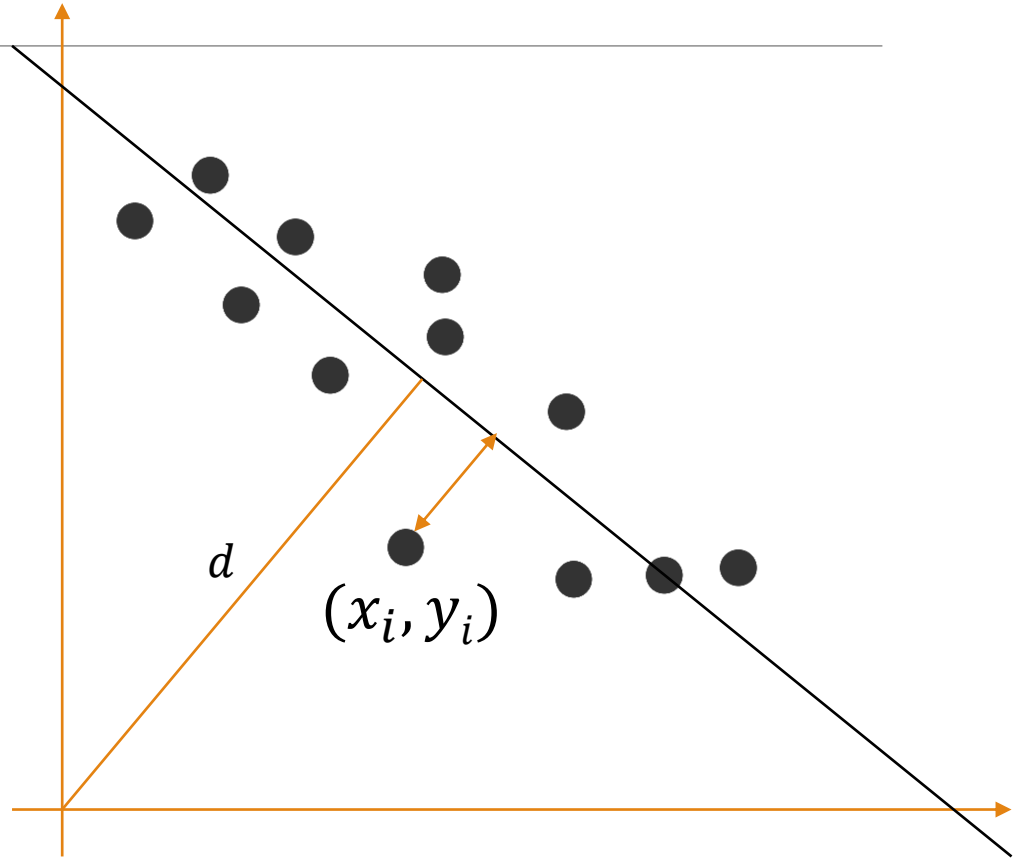


Number of non-zero coefficients (Linear Regression): 50
Number of non-zero coefficients (Lasso Regression): 16
True number of non-zero coefficients =10 and total number of coefficients =50

Total least squares (TLS)

- Equation of a line can be written as $ax + by = d$
- $a^2 + b^2 = 1$, where (a, b) is the unit normal to the line
- Distance between line and the origin is given as d
- Extension to 3d, $ax + by + cz = d$ and $a^2 + b^2 + c^2 = 1$
- Distance (perpendicular) from point to line is $\frac{|ax+by-d|}{\sqrt{a^2+b^2}}$
- Loss function:

$$E = \sum_{i=1}^{n} (ax_i + by_i - d)^2$$



$\hat{a}, \hat{b}, \hat{d}, = \underset{a,b,d}{\arg\min}\ E$

$\hat{a}, \hat{b}, \hat{d}, = \underset{a,b,d}{\arg\min}\ E$ such that $a^2 + b^2 = 1$

Distance from a point to a line - Wikipedia

Content adapted from Fitting-Svetlana Lazebnik

Total least squares (TLS)

- Solve for d ➔

$$\frac{\partial E}{\partial d} = 0$$

$$E = \sum_{i=1}^{n} (ax_i + by_i - d)^2$$

$$\sum_{i=1}^{n} 2\,(ax_i + by_i - d)\;\;(-1) = 0$$

$$\sum_{i=1}^{n} (-ax_i - byi + d) = 0$$

$$n\,d = \sum_{i=1}^{n} (ax_i + byi)$$

$$d = \frac{a}{n}\sum_{i=1}^{n} x_i + \frac{b}{n}\sum_{i=1}^{n} y_i = a\bar{x} + b\bar{y}$$

$$E = \sum_{i=1}^{n} (a(x_i - \bar{x}) + b(y_i - \bar{y}))^2 = \left\| \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \right\|^2 = \|\boldsymbol{Xu}\|^2$$

Total least squares (TLS)

- Optimization problem minimizes $\|\boldsymbol{X}\boldsymbol{u}\|^2$ subject to $\|\boldsymbol{u}\|^2 = 1$
- Lagrangian function E($\boldsymbol{u}$, λ)= $\|\boldsymbol{X}\boldsymbol{u}\|^2 + \lambda\ (\|\boldsymbol{u}\|^2\text{-}1)$ and λ is Lagrange multiplier
- $\frac{\partial E}{\partial \boldsymbol{u}} = 0, \Rightarrow 2\ \boldsymbol{X}^{\mathsf{T}}(\boldsymbol{X}\ \boldsymbol{u}) + \lambda\ \boldsymbol{u} = 0$ ➔ $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\ \boldsymbol{u} = -\lambda\ \boldsymbol{u}$
- $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\ \boldsymbol{u} = -\lambda\ \boldsymbol{u}$ Some thing familiar?
- $\boldsymbol{u}$ is the eigenvector of $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})$, if $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})$ is p by p vector and there can be p number of eigen vectors and eigen values (λ)
- Which one should I pick?
- $\frac{\partial E}{\partial \lambda} = 0$ ➔ $\|\boldsymbol{u}\|^2\text{-}1$ => $\|\boldsymbol{u}\|^2 = 1$ already known.
- $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\ \boldsymbol{u} = -\lambda\ \boldsymbol{u}$ multiply both sides by $\boldsymbol{u}^{\mathsf{T}}$
- $\boldsymbol{u}^{\mathsf{T}}\ (\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\ \boldsymbol{u} = -\lambda\ \boldsymbol{u}^{\mathsf{T}}\boldsymbol{u}$
- $(\boldsymbol{u}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}})\ \ (\boldsymbol{X}\ \boldsymbol{u}) = -\lambda$ any insight for picking value to value for λ ?

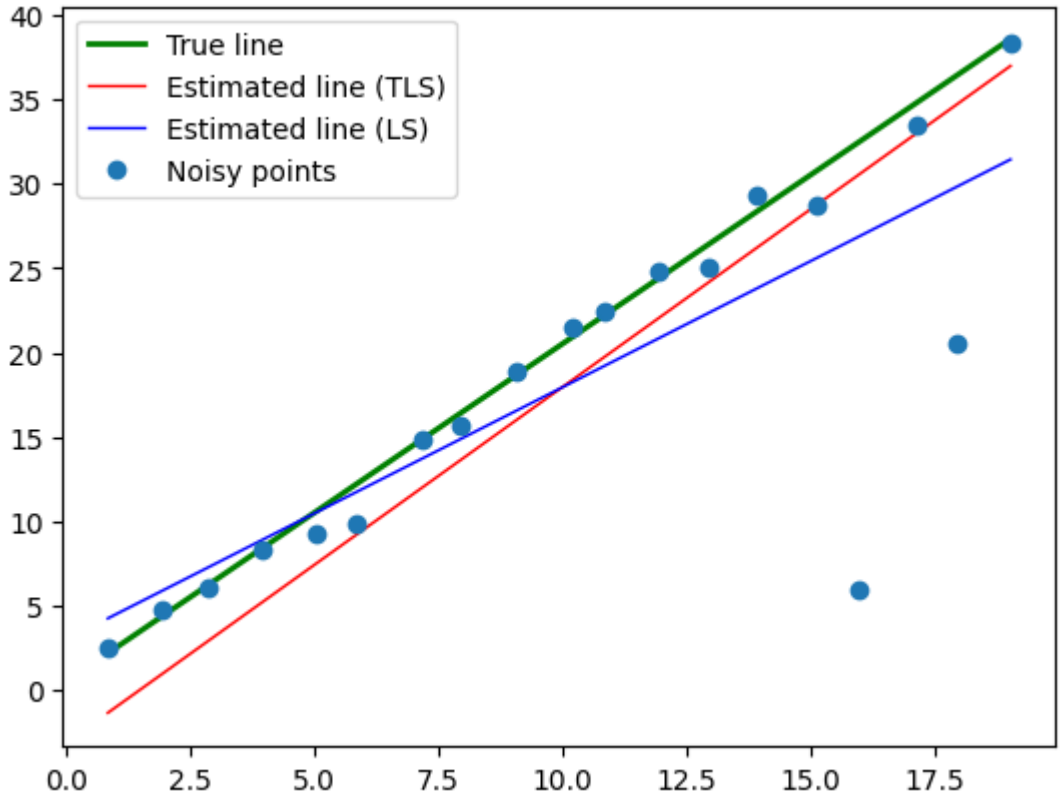
Total least squares (TLS)

- $(\boldsymbol{u}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}})\ (\boldsymbol{X}\,\boldsymbol{u}) = -\lambda$ any insight for picking value to value for λ ?
- $\|\boldsymbol{X}\boldsymbol{u}\|^2 = -\lambda$, we need to minimize the $\|\mathbf{X}\mathbf{u}\|^2$.
- Pick a smallest eigen value.
- Solution: eigenvector of $(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})$ associated with the smallest eigenvalue.

  e. g. , $\begin{bmatrix} 0.4 & 0.68 \\ 0.05 & 0.39 \end{bmatrix}$ for this matrix eigenvalues and eigenvectors are $[0.58\ 0.22]$ and $\begin{bmatrix} 0.96 & -0.96 \\ 0.25 & 0.26 \end{bmatrix}$

- LS assume that measurement errors primarily affect the dependent variable (the y-values).
- TLS consider both dependent and independent variables have measurement errors.
- TLS is less sensitive to outliers compared to traditional least squares.

TLS and LS


True line
Estimated line (TLS)
Estimated line (LS)
Noisy points
40
35
30
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

Self Study

- **RANSAC** (RANdom SAmple Consensus)

  - Fischler, Martin A., and Robert C. Bolles. "Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography." *Communications of the ACM* 24.6 (1981): 381-395.

  - https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.RANSACRegressor.html

  - Robust linear model estimation using RANSAC — scikit-learn 1.3.0 documentation





Thank You

Q & A